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BLACK HOLES AS SPECTRAL FILTERS: RESONANCES AND THE ROLE OF INITIAL DATA

Jose Antonio León Vega

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Key words: black holes, quasinormal modes, waveform, excitation, resonances, suppression, tail.

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Abstract

Black-hole ringdown is commonly described as a superposition of quasinormal modes, whose complex frequencies are fixed by the background spacetime. However, the amplitudes with which these modes are excited depend on the perturbation that triggers the response. This work studies this dependence in the framework of linear perturbations of a Schwarzschild black hole, focusing on axial Regge–Wheeler perturbations.

In the Green-function formulation, ringdown appears as the contribution from the quasinormal-mode poles in the complex frequency plane. The excitation coefficient of each mode factorizes into a spacetime-dependent excitation factor and a source-dependent overlap. This overlap can be understood as a weighted Fourier transform of the initial data evaluated at the quasinormal-mode frequency. In this sense, the black hole acts as a spectral filter: modes are enhanced when the perturbation contains spectral power near their characteristic frequencies and are suppressed otherwise.

To make this mechanism explicit, localized oscillatory Gaussian initial data are introduced, with width σ and carrier frequency ν . These parameters allow independent control of the spectral bandwidth and dominant frequency of the perturbation. Analytical estimates based on asymptotic overlaps are compared with numerical time-domain evolutions, showing that the excitation landscape is governed by the spectral content of the initial data. Finally, the late-time power-law tail is analyzed. The same spectral parameter $\alpha = \sigma\nu$ that controls the transition from broadband to frequency-selective excitation also produces an exponential suppression of the tail. This provides a unified picture of black-hole ringdown as a spectral filtering process and offers an initial physical interpretation of the suppression mechanism of late-time tails in perturbation theory.

Resum

El ringdown d'un forat negre es descriu habitualment com una superposició de modes quasinormals, les freqüències complexes dels quals estan determinades per l'espai-temps de fons. Tanmateix, les amplituds amb què aquests modes són excitats depenen de la pertorbació que genera la resposta. Aquest treball estudia aquesta dependència dins el marc de les pertorbacions lineals d'un forat negre de Schwarzschild, centrant-se en pertorbacions axials de Regge–Wheeler.

En la formulació mitjançant la funció de Green, el ringdown apareix com la contribució dels pols dels modes quasinormals en el pla complex de freqüències. El coeficient d'excitació de cada mode es factoritza en un factor d'excitació dependent de l'espai-temps i un solapament dependent de la font. Aquest solapament es pot entendre com una transformada de Fourier ponderada de les dades inicials avaluada a la freqüència del mode quasinormal. En aquest sentit, el forat negre actua com un filtre espectral: els modes s'amplifiquen quan la pertorbació conté potència espectral prop de les seves freqüències característiques i se suprimeixen en cas contrari.

Per fer explícit aquest mecanisme, s'introdueixen dades inicials gaussianes oscil·latòries localitzades, amb amplada σ i freqüència portadora ν . Aquests paràmetres permeten controlar de manera independent l'amplada espectral i la freqüència dominant de la pertorbació. Les estimacions analítiques basades en solapaments asimptòtics es comparen amb evolucions numèriques en el domini temporal, mostrant que l'estructura d'excitació està governada pel contingut espectral de les dades inicials. Finalment, s'analitza la cua tardana de tipus llei de potència. El mateix paràmetre espectral $\alpha = \sigma\nu$ que controla la transició entre una excitació de banda ampla i una excitació selectiva en freqüència produeix també una supressió exponencial de la cua. Això proporciona una visió unificada del ringdown dels forats negres com un procés de filtratge espectral i ofereix una primera interpretació física del mecanisme de supressió de les cues tardanes en teoria de pertorbacions.

Resumen

El ringdown de un agujero negro se describe habitualmente como una superposición de modos cuasinormales, cuyas frecuencias complejas están determinadas por el espaciotiempo de fondo. Sin embargo, las amplitudes con las que estos modos se excitan dependen de la perturbación que genera la respuesta. Este trabajo estudia dicha dependencia en el marco de las perturbaciones lineales de un agujero negro de Schwarzschild, centrándose en perturbaciones axiales de Regge–Wheeler.

En la formulación mediante la función de Green, el ringdown aparece como la contribución de los polos cuasinormales en el plano complejo de frecuencias. El coeficiente de excitación de cada modo se factoriza en un factor de excitación dependiente del espaciotiempo y un solapamiento dependiente de la fuente. Este solapamiento puede entenderse como una transformada de Fourier ponderada de los datos iniciales evaluada en la frecuencia del modo cuasinormal. En este sentido, el agujero negro actúa como un filtro espectral: los modos se amplifican cuando la perturbación contiene potencia espectral cerca de sus frecuencias características y se suprimen en caso contrario.

Para hacer explícito este mecanismo, se introducen datos iniciales gaussianos oscilatorios localizados, con anchura σ y frecuencia portadora ν . Estos parámetros permiten controlar de manera independiente el ancho espectral y la frecuencia dominante de la perturbación. Las estimaciones analíticas basadas en solapamientos asintóticos se comparan con evoluciones numéricas en el dominio temporal, mostrando que la estructura de excitación está gobernada por el contenido espectral de los datos iniciales. Finalmente, se analiza la cola tardía de tipo ley de potencias. El mismo parámetro espectral $\alpha = \sigma\nu$ que controla la transición entre una excitación de banda ancha y una excitación selectiva en frecuencia produce también una supresión exponencial de la cola. Esto proporciona una visión unificada del ringdown de agujeros negros como un proceso de filtrado espectral y ofrece una primera interpretación física del mecanismo de supresión de las colas tardías en teoría de perturbaciones.

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Black holes and gravitational-wave ringdown

1.1 Spacetime geometry and gravitational dynamics

General Relativity (GR), formulated by Einstein in 1915 [1], describes gravity not as a force in the Newtonian sense, but as a manifestation of spacetime curvature [2–5]. In this framework, freely falling test particles follow geodesics of the spacetime manifold, while matter and energy determine its geometry through the Einstein field equations.

Spacetime is modeled as a differentiable manifold endowed with a Lorentzian metric tensor $g_{\mu\nu}$, which determines distances, causal structure, and local geometry through the line element

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu. \quad (1.1)$$

The dynamics of the gravitational field are governed by the Einstein field equations,

$$G_{\mu\nu} = 8\pi T_{\mu\nu}, \quad (1.2)$$

where $G_{\mu\nu}$ is the Einstein tensor, $T_{\mu\nu}$ is the stress–energy tensor, and geometrized units with $G = c = M = 1$ are used throughout.

In vacuum, where $T_{\mu\nu} = 0$, Eq. (1.2) reduces to

$$R_{\mu\nu} = 0, \quad (1.3)$$

whose exact solutions include the Schwarzschild and Kerr black holes. These solutions play a central role in gravitational physics, since they describe the exterior spacetime of compact objects after gravitational collapse.

Among them, the Schwarzschild geometry is the simplest black-hole solution: it is static, spherically symmetric, and uncharged. Because of this simplicity, it provides the natural starting point for black-hole perturbation theory and for the study of quasinormal ringing. In the present work, it will serve as the reference background on which linear perturbations are introduced and evolved.

1.2 Stationary black-hole solutions

In vacuum, the Einstein field equations admit exact solutions describing self-consistent curved spacetimes in the absence of matter sources. Among them, black-hole solutions are of particular importance. They are characterized by the presence of an event horizon, which separates the region causally connected to future null infinity from that from which no signal can escape [6].

Under the assumptions of stationarity and asymptotic flatness, the family of analytic black-hole solutions in four-dimensional electrovacuum General Relativity is remarkably restricted. In essence, such solutions are completely characterized by a small set of global parameters: the mass M , the angular momentum J , and the electric charge Q [6]. The corresponding stationary black-hole geometries are summarized in Figure 1.1.

In the present work we focus on the Schwarzschild solution, which describes a static, spherically symmetric, uncharged black hole. Its simplicity makes it the natural starting point for the study of linear perturbations.

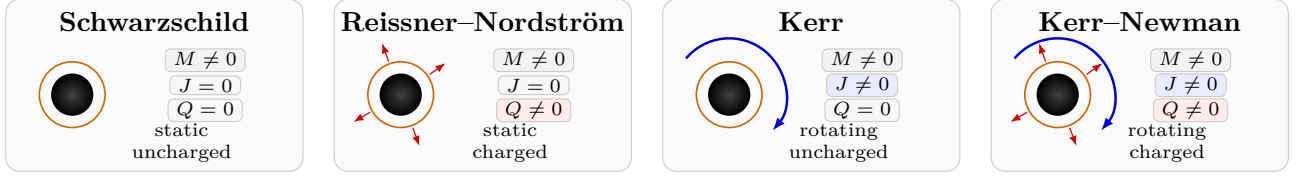


Figure 1.1: Stationary black-hole families in four-dimensional Einstein–Maxwell theory.

1.3 Linearized gravity: from Minkowski waves to black-hole perturbations

Gravitational Waves (GWs) are among the most important predictions of GR [5, 7]. They are propagating disturbances of the spacetime geometry, generated by accelerated mass distributions and carrying energy and angular momentum. Because the Einstein field equations are highly nonlinear, an exact treatment is generally difficult. In the weak-field regime, however, the theory can be consistently linearized, providing both the simplest description of gravitational radiation and the conceptual starting point for black-hole perturbation theory [7, 8].

The physical reality of GWs, debated in the early history of GR, was clarified at the 1957 Chapel Hill conference, notably through Feynman’s sticky-bead argument that gravitational waves can have observable effects and carry energy [9, 10]. Their existence was later supported indirectly through the orbital decay of the Hulse–Taylor binary pulsar [11], and finally established directly by the first Laser Interferometer Gravitational-Wave Observatory (LIGO) detection in 2015 [12].

1.3.1 Linearization around Minkowski spacetime

In the weak-field limit, the spacetime metric may be written as a small perturbation of flat Minkowski spacetime,

$$g_{\mu\nu} = \eta_{\mu\nu} + \epsilon h_{\mu\nu}, \quad \epsilon \ll 1, \quad (1.4)$$

where $\eta_{\mu\nu} = \text{diag}(-1, 1, 1, 1)$. At first order in $h_{\mu\nu}$, indices are raised and lowered with $\eta_{\mu\nu}$, and the Einstein equations reduce to a linear system for the perturbation.

A convenient variable is the trace-reversed perturbation,

$$\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h, \quad h \equiv \eta^{\alpha\beta} h_{\alpha\beta}, \quad (1.5)$$

together with the harmonic gauge condition $\partial^\nu \bar{h}_{\mu\nu} = 0$ [5]. In this gauge, the linearized Einstein equations take the wave-like form

$$\square \bar{h}_{\mu\nu} = -16\pi T_{\mu\nu}, \quad \square \equiv \eta^{\alpha\beta} \partial_\alpha \partial_\beta. \quad (1.6)$$

In vacuum, $T_{\mu\nu} = 0$, so that

$$\square \bar{h}_{\mu\nu} = 0, \quad (1.7)$$

whose simplest solutions are plane waves,

$$\bar{h}_{\mu\nu}(x) = \Re(A_{\mu\nu} e^{ik_\alpha x^\alpha}), \quad k^\alpha k_\alpha = 0. \quad (1.8)$$

The null condition implies that gravitational waves propagate at the speed of light. After further gauge fixing, one reaches the Transverse–Traceless (TT) gauge, in which only two independent polarizations remain, conventionally denoted by $+$ and $\times$.

Linearization is not restricted to flat spacetime. More generally, one may perturb an exact background solution $g_{\mu\nu}^{(0)}$ according to

$$g_{\mu\nu} = g_{\mu\nu}^{(0)} + \epsilon h_{\mu\nu}, \quad \epsilon \ll 1 \quad (1.9)$$

When $g_{\mu\nu}^{(0)} = \eta_{\mu\nu}$, one recovers the flat-spacetime description above. When the background is curved, the perturbations couple to the geometry and the problem becomes one of wave propagation on a nontrivial spacetime. This is precisely the setting of black-hole perturbation theory. In the Schwarzschild case, spherical symmetry allows a decomposition into odd-parity (axial) and even-parity (polar) sectors, governed by the Regge–Wheeler and Zerilli equations, respectively. For rotating black holes, the corresponding framework is provided by the Teukolsky formalism.

1.3.2 Gravitational-wave signals and the relevance of ringdown

Astrophysical GW signals from compact-binary coalescences are commonly divided into three stages: inspiral, merger, and ringdown. During the inspiral, the binary loses energy and angular momentum through gravitational radiation, producing the characteristic chirp signal. As the separation decreases, nonlinear effects become dominant and the system enters the merger phase, which generally requires full Numerical Relativity (NR). After merger, the remnant relaxes toward a stationary black-hole geometry through a sequence of damped oscillations known as ringdown.

In this final stage, linear perturbation theory becomes accurate again, and the waveform can be described as a superposition of Quasi-Normal Modes (QNMs) [13],

$$\Psi_{\ell m}(t) \sim \sum_{n=0}^{\infty} A_{\ell mn} e^{-i\omega_{\ell mn}(t-t_0)}, \quad \omega_{\ell mn} = \omega_{\ell mn}^R - i\omega_{\ell mn}^I = \omega_{\ell mn}^R - \frac{i}{\tau_{\ell mn}}, \quad \omega_{\ell mn}^I > 0. \quad (1.10)$$

Here $\omega_{\ell mn}^R$ is the oscillation frequency, $\omega_{\ell mn}^I$ is the damping rate, and $\tau_{\ell mn} = 1/\omega_{\ell mn}^I$ is the corresponding damping time. The coefficient $A_{\ell mn}$ is in general complex. Therefore, if one writes $A_{\ell mn} = |A_{\ell mn}|e^{i\phi_{\ell mn}}$, the phase $\phi_{\ell mn}$ is already included in $A_{\ell mn}$. More generally, constant phase factors associated with the choice of time origin, waveform convention, or mode normalization can be absorbed into this complex amplitude.

The indices (ℓ, m, n) label the angular and radial structure of the mode. The integer ℓ is the multipole order, m is the azimuthal number with $-\ell \leq m \leq \ell$, and $n = 0, 1, 2, \dots$ labels the overtone, with $n = 0$ denoting the fundamental mode and larger n corresponding, in general, to more strongly damped overtones. For a rotating Kerr black hole the frequencies depend on the final mass and spin and on the full set (ℓ, m, n) . In the Schwarzschild limit, however, spherical symmetry makes the spectrum degenerate in m , so that one may write $\omega_{\ell mn} \equiv \omega_{\ell n}$. Since gravitational radiation starts at quadrupolar order, the dominant Schwarzschild sector is usually the fundamental $\ell = 2, n = 0$ mode, which will be the main focus of this work.

The ringdown phase is especially relevant for gravitational-wave astronomy because it provides a direct connection between black-hole perturbation theory and observations. Measurements of ringdown frequencies and damping times make it possible to infer the properties of the remnant and to test the consistency of GR in the strong-field regime. In this sense, ringdown constitutes the theoretical basis of black-hole spectroscopy.

However, the knowledge of the Quasi-normal Mode (QNM) spectrum alone is not sufficient. One must also understand how the different modes are excited, which modes dominate the signal, and how the observed waveform depends on the initial perturbation. These questions motivate the study

of excitation coefficients, asymptotic reconstructions, and the dependence of the waveform on the spectral content of the perturbation. Although astrophysical remnants are generically rotating [14] and therefore described by Kerr geometry, the Schwarzschild case provides the simplest framework in which these mechanisms can be analyzed with conceptual clarity.

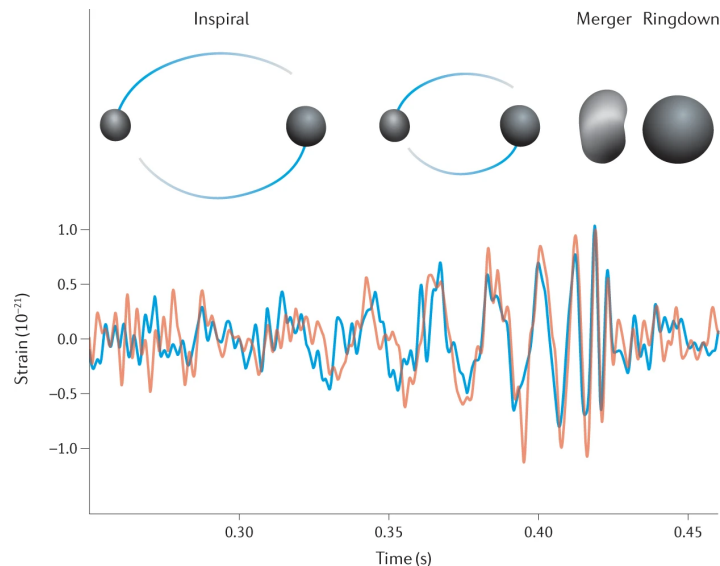


Figure 1.2: **Inspiral, merger and ringdown.** NR simulations of a binary Black Hole (BH) coalescence (top) and the corresponding GW strain observed at Earth (bottom). The signal exhibits three phases: inspiral, merger and ringdown. The last one is dominated by the QNMs of the remnant black hole [15].

1.4 Schwarzschild spacetime and linear perturbations

Among the stationary black-hole solutions of General Relativity, the Schwarzschild geometry is the simplest and historically the first to be discovered [16]. It describes the exterior spacetime of a static, spherically symmetric, uncharged body, and constitutes the unique asymptotically flat, spherically symmetric vacuum solution of the Einstein field equations. In Schwarzschild coordinates, the line element is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \left(1 - \frac{2M}{r}\right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\varphi^2). \quad (1.11)$$

The surface $r = 2M$ is the event horizon, while $r = 0$ is a genuine curvature singularity [8]. For later use, it is convenient to introduce the tortoise coordinate r_* , defined by

$$\frac{dr_*}{dr} = \left(1 - \frac{2M}{r}\right)^{-1}, \quad r_* = r + 2M \ln\left(\frac{r}{2M} - 1\right). \quad (1.12)$$

This coordinate maps the horizon to $r_* \rightarrow -\infty$ and spatial infinity to $r_* \rightarrow +\infty$, and it will play a central role in black-hole perturbation theory.

An important geometric feature of Schwarzschild spacetime is the photon sphere at

$$r = 3M, \quad (1.13)$$

which corresponds to an unstable circular null orbit. This region is closely related to the peak of the effective potentials governing linear perturbations, and therefore to the scattering and ringdown

properties of the black hole. In particular, in the eikonal limit $\ell \gg 1$, perturbations behave in close analogy with null rays trapped near the photon sphere. In this regime, the peak of the effective potential approaches the photon sphere and the oscillation frequency of the ringdown is associated with the orbital frequency of the unstable null orbit, while its decay is associated with the instability timescale of that orbit [17].

Linear perturbations of Schwarzschild spacetime:

We now consider small gravitational perturbations of the Schwarzschild metric ($s = \pm 2$), writing the full spacetime using Eq. (1.9)

$$g_{\mu\nu} = g_{\mu\nu}^{\text{Schw}} + \epsilon h_{\mu\nu}, \quad \epsilon \ll 1. \quad (1.14)$$

Substituting this decomposition into the Einstein field equations and expanding to first order in $h_{\mu\nu}$, one obtains the linearized equations governing the propagation of perturbations on the Schwarzschild background.

A major simplification follows from the spherical symmetry of the Schwarzschild background. The perturbations can be decomposed into tensor spherical harmonics labelled by (ℓ, m) , so that the different angular components decouple at linear order. Since the background is invariant under rotations, the radial operators and the effective potentials depend on ℓ but not on m . Thus the different azimuthal components with fixed ℓ are degenerate and satisfy the same one-dimensional master equation.

Moreover, under the parity transformation

$$(\theta, \varphi) \mapsto (\pi - \theta, \varphi + \pi),$$

the perturbations split into odd-parity (axial) and even-parity (polar) sectors. For radiative modes $\ell \geq 2$, both sectors can be described in terms of master functions $\Psi_{\ell m}^{(\pm)}$ [18] satisfying

$$\left[-\frac{\partial^2}{\partial t^2} + \frac{\partial^2}{\partial r_*^2} - V_\ell^{(\pm)}(r) \right] \Psi_{\ell m}^{(\pm)}(t, r) = 0. \quad (1.15)$$

Here the minus sign refers to the odd-parity Regge–Wheeler (RW) sector, while the plus sign refers to the even-parity (Zerilli) one.

For odd parity, the effective potential is the Regge–Wheeler potential,

$$V_\ell^{(-)}(r) = \left(1 - \frac{2M}{r} \right) \left[\frac{\ell(\ell+1)}{r^2} - \frac{6M}{r^3} \right], \quad (1.16)$$

whereas for even parity one obtains the Zerilli potential,

$$V_\ell^{(+)}(r) = \left(1 - \frac{2M}{r} \right) \frac{2\lambda^2(\lambda+1)r^3 + 6\lambda^2Mr^2 + 18\lambda M^2r + 18M^3}{r^3(\lambda r + 3M)^2}, \quad \lambda \equiv \frac{1}{2}(\ell-1)(\ell+2). \quad (1.17)$$

Thus, linear perturbations of Schwarzschild reduce to a one-dimensional scattering problem in the tortoise coordinate. The effective potentials vanish both at the horizon and at spatial infinity, while developing a localized barrier in the intermediate region, close to the photon sphere. This structure underlies the main phenomena of black-hole perturbation theory: wave scattering, quasinormal ringing, and late-time tails.

Although the Regge–Wheeler and Zerilli equations describe perturbations of different parity, in vacuum they share the same quasinormal-mode spectrum. This phenomenon is called *isospectrality*.

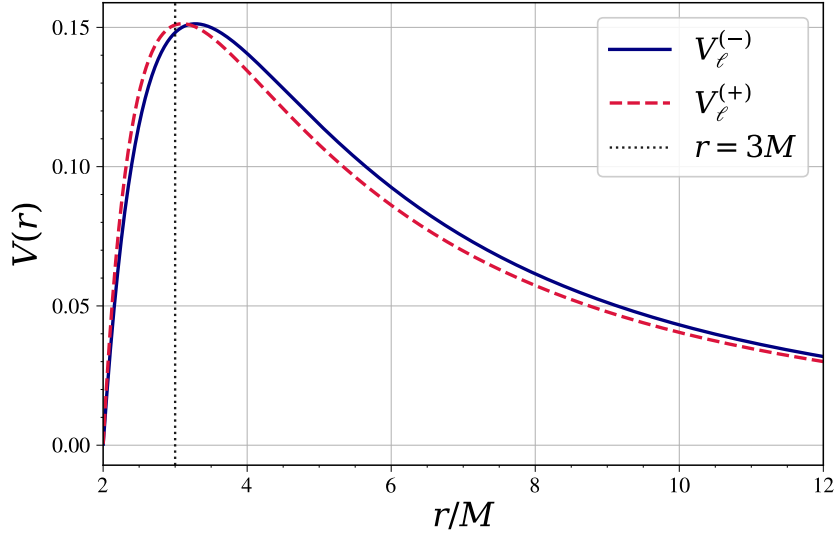


Figure 1.3: **Comparison between the RW and Zerilli potentials for gravitational perturbations and $\ell = 2$.** Although the two potentials are not identical, they display a very similar barrier structure and, in vacuum, lead to the same quasinormal-mode spectrum.

In the present work, special attention will be paid to the Regge–Wheeler framework, which provides a particularly clean setting for the study of mode excitation, asymptotic reconstruction, and the dependence of the response on the spectral content of the initial data.

The choice $\ell = 2$ is not arbitrary. In gravitational radiation, the lowest radiative multipole is the quadrupole, $\ell = 2$, while the $\ell = 0$ and $\ell = 1$ sectors are non-radiative in vacuum and are associated with changes in the background parameters or gauge degrees of freedom. For this reason, the $\ell = 2$ sector gives the leading contribution to the Schwarzschild gravitational response and is usually the dominant one in the ringdown signal. In the numerical applications of this work we will therefore focus on axial Regge–Wheeler perturbations with $\ell = 2$, while the preceding equations remain valid for any radiative multipole $\ell \geq 2$.

Since the Regge–Wheeler and Zerilli potentials vanish both at the horizon, $r_* \rightarrow -\infty$, and at spatial infinity, $r_* \rightarrow +\infty$, the master equation reduces in both limits to the free wave equation. With the convention $e^{-i\omega t}$, the two independent asymptotic behaviours are

$$\hat{\Psi}(r_*) \sim A_{\text{out}} e^{+i\omega r_*} + A_{\text{in}} e^{-i\omega r_*}. \quad (1.18)$$

The first term represents radiation outgoing towards spatial infinity, while the second represents radiation ingoing from spatial infinity [19]. Analogously, at the horizon one imposes purely ingoing behaviour.

These radiative boundary conditions turn the Schwarzschild perturbation problem into an open scattering problem. Quasinormal modes are then selected as the solutions satisfying the appropriate outgoing condition at infinity and ingoing condition at the horizon. This observation provides the starting point for the Green-function formulation developed in Chapter 2, where the ringdown signal will be identified with the pole contribution of the black-hole response.

Green-function formulation of quasinormal modes

This chapter summarizes the Green-function framework underlying the quasinormal-mode description of Schwarzschild black-hole ringdown. Starting from the Regge–Wheeler equation, we introduce the frequency-domain problem, construct the corresponding Green function, and identify quasinormal modes as the poles of its analytic continuation. The technical derivation of the Green function, the jump conditions, the residue calculation, and the reality symmetry of the pole pairs are collected in Appendix B.

2.1 Frequency-domain problem and radiative solutions

Axial gravitational perturbations of the Schwarzschild spacetime satisfy the Regge–Wheeler equation

$$\left[\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial r_*^2} + V(r) \right] \Psi(t, r_*) = \mathcal{S}(t, r_*), \quad (2.1)$$

where r_* is the tortoise coordinate. We introduce the unilateral Laplace transform and then set

$$s = -i\omega, \quad \hat{\Psi}(\omega, r_*) \equiv \tilde{\Psi}(-i\omega, r_*). \quad (2.2)$$

The transformed Regge–Wheeler equation becomes

$$\frac{d^2 \hat{\Psi}}{dr_*^2} + [\omega^2 - V(r)] \hat{\Psi} = I(\omega, r_*), \quad (2.3)$$

with

$$I(\omega, r_*) = i\omega f(r_*) - g(r_*) - \tilde{\mathcal{S}}(-i\omega, r_*), \quad (2.4)$$

where

$$f(r_*) = \Psi(0, r_*), \quad g(r_*) = \partial_t \Psi(0, r_*). \quad (2.5)$$

In the source-free case, $\mathcal{S} = 0$, this reduces to

$$I(\omega, r_*) = i\omega f(r_*) - g(r_*). \quad (2.6)$$

The associated homogeneous equation is

$$\frac{d^2 \psi}{dr_*^2} + [\omega^2 - V(r)] \psi = 0. \quad (2.7)$$

Since $V(r) \rightarrow 0$ as $r_* \rightarrow \pm\infty$, the asymptotic solutions are plane waves. With the convention $e^{-i\omega t}$, the radiative boundary conditions are

$$\psi \sim e^{-i\omega r_*} \quad (r_* \rightarrow -\infty), \quad \psi \sim e^{+i\omega r_*} \quad (r_* \rightarrow +\infty), \quad (2.8)$$

corresponding respectively to a purely ingoing wave at the horizon and a purely outgoing wave at infinity.

We therefore introduce two distinguished homogeneous solutions [20]. The first one, ψ_{in} , is normalized to be purely ingoing at the horizon:

$$\psi_{\text{in}}(r_*, \omega) \sim \begin{cases} e^{-i\omega r_*}, & r_* \rightarrow -\infty, \\ A_{\text{in}}(\omega)e^{-i\omega r_*} + A_{\text{out}}(\omega)e^{+i\omega r_*}, & r_* \rightarrow +\infty. \end{cases} \quad (2.9)$$

The second one, ψ_{up} , is normalized to be purely outgoing at spatial infinity:

$$\psi_{\text{up}}(r_*, \omega) \sim \begin{cases} B_{\text{in}}(\omega)e^{-i\omega r_*} + B_{\text{out}}(\omega)e^{+i\omega r_*}, & r_* \rightarrow -\infty, \\ e^{+i\omega r_*}, & r_* \rightarrow +\infty. \end{cases} \quad (2.10)$$

Their Wronskian,

$$W(\omega) \equiv \psi_{\text{in}}\psi'_{\text{up}} - \psi_{\text{up}}\psi'_{\text{in}}, \quad (2.11)$$

is independent of r_* . Evaluating it at spatial infinity gives

$$W(\omega) = 2i\omega A_{\text{in}}(\omega), \quad (2.12)$$

while evaluating it at the horizon gives $A_{\text{in}}(\omega) = B_{\text{out}}(\omega)$.

2.2 Green function and quasinormal poles

The frequency-domain Green function [20] is defined by

$$\left[\frac{d^2}{dr_*^2} + \omega^2 - V(r) \right] \hat{G}(r_*, r'_*; \omega) = \delta(r_* - r'_*). \quad (2.13)$$

The boundary conditions select ψ_{in} to the left of the source point and ψ_{up} to its right. Matching the two pieces at $r_* = r'_*$ gives the standard Green function

$$\hat{G}(r_*, r'_*; \omega) = \frac{\psi_{\text{in}}(r_<, \omega) \psi_{\text{up}}(r_>, \omega)}{W(\omega)}, \quad r_< \equiv \min(r_*, r'_*), \quad r_> \equiv \max(r_*, r'_*). \quad (2.14)$$

The solution of the inhomogeneous problem is therefore

$$\hat{\Psi}(r_*, \omega) = \int_{-\infty}^{+\infty} \hat{G}(r_*, r'_*; \omega) I(\omega, r'_*) dr'_*. \quad (2.15)$$

Quasinormal modes are selected by imposing the radiative boundary conditions at both asymptotic ends simultaneously [13, 20]. Since ψ_{in} is already purely ingoing at the horizon, it becomes a quasinormal mode when the incoming component at infinity vanishes:

$$A_{\text{in}}(\omega_n) = 0. \quad (2.16)$$

Using Eq. (2.12), this condition is equivalent to $W(\omega_n) = 0$. Thus the quasinormal frequencies are precisely the poles of the analytically continued Green function. With the convention $e^{-i\omega t}$, the physical modes lie in the lower half-plane:

$$\omega_n = \omega_n^{\text{R}} - i\omega_n^{\text{I}}, \quad \omega_n^{\text{I}} > 0. \quad (2.17)$$

A schematic contour deformation is shown in Fig. 2.1. The pole sum gives the quasinormal contribution, while the branch cut, omitted in the figure, gives the late-time tail.

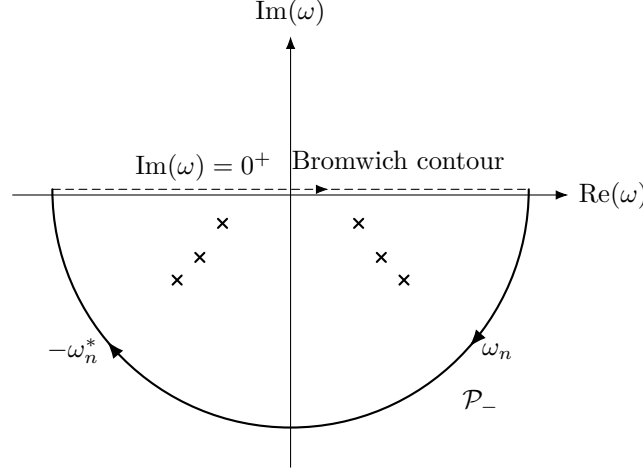


Figure 2.1: Schematic contour deformation used to isolate the quasinormal-mode contribution to the inverse transform. Under $s = -i\omega$, the Bromwich line maps to a horizontal contour in the complex ω -plane, which is closed clockwise in the lower half-plane for $t > 0$. The branch-cut contribution associated with the late-time tail is not shown.

2.3 Ringdown expansion and excitation coefficients

The inverse transform is evaluated by deforming the contour into the lower half-plane for $t > 0$. The contour encloses the quasinormal poles and, in Schwarzschild, also a branch cut associated with the late-time tail. In the ringdown regime we keep only the pole contribution [13, 20]:

$$\Psi_{\text{QNM}}(t, r_*) = -i \sum_n e^{-i\omega_n t} \text{Res}_{\omega=\omega_n} \hat{\Psi}(r_*, \omega). \quad (2.18)$$

For simple poles, using Eq. (2.15) gives

$$\Psi_{\text{QNM}}(t, r_*) = -i \sum_n e^{-i\omega_n t} \int_{-\infty}^{+\infty} I(\omega_n, r'_*) \frac{\psi_{\text{in}}(r_<, \omega_n) \psi_{\text{up}}(r_>, \omega_n)}{W'(\omega_n)} dr'_*. \quad (2.19)$$

This expression is still exact within the pole approximation: no far-zone approximation has yet been introduced.

We now specialize to an observer in the asymptotic wave zone, $r_* \rightarrow +\infty$. For localized initial data or source support, $r_> = r_*$ and $r_< = r'_*$ throughout the integral. Since $\psi_{\text{up}}(r_*, \omega_n) \sim e^{+i\omega_n r_*}$, it is natural to introduce the retarded time

$$u \equiv t - r_*. \quad (2.20)$$

Then

$$\Psi_{\text{QNM}}(u) = -i \sum_{\omega_n \in \mathcal{P}_-} e^{-i\omega_n u} \frac{1}{W'(\omega_n)} \int_{-\infty}^{+\infty} \psi_{\text{in}}(r'_*, \omega_n) I(\omega_n, r'_*) dr'_*, \quad (2.21)$$

where $\mathcal{P}_-$ denotes the set of quasinormal poles in the lower half-plane.

Differentiating Eq. (2.12) and using $A_{\text{in}}(\omega_n) = 0$, one obtains

$$W'(\omega_n) = 2i\omega_n A'_{\text{in}}(\omega_n). \quad (2.22)$$

We then define the excitation factor as

$$B_n \equiv i \frac{A_{\text{out}}(\omega_n)}{W'(\omega_n)} = \frac{A_{\text{out}}(\omega_n)}{2\omega_n A'_{\text{in}}(\omega_n)}. \quad (2.23)$$

The source-dependent overlap is

$$\mathcal{T}_n \equiv \int_{-\infty}^{+\infty} \frac{\psi_{\text{in}}(r'_*, \omega_n) I(\omega_n, r'_*)}{A_{\text{out}}(\omega_n)} dr'_*, \quad (2.24)$$

and the excitation coefficient is

$$\mathcal{C}_n = B_n \mathcal{T}_n. \quad (2.25)$$

Eq. (2.25) shows that modal amplitude naturally separates into a black-hole background factor B_n and a source-dependent factor $\mathcal{T}_n$. Therefore the wave-zone pole contribution can be written as

$$\Psi_{\text{QNM}}(u) = - \sum_{\omega_n \in \mathcal{P}_-} \mathcal{C}_n e^{-i\omega_n u}. \quad (2.26)$$

For real initial data and a real Schwarzschild potential, the negative-frequency poles are the mirror partners of the positive-frequency ones, $\bar{\omega}_n = -\omega_n^*$, with conjugate modal amplitudes. Therefore the sum over both branches can be written as a sum over the positive-frequency branch only, [13, 21]

$$\Psi_{\text{QNM}}(u) = -2 \operatorname{Re} \sum_{\operatorname{Re}(\omega_n) > 0} \mathcal{C}_n e^{-i\omega_n u}. \quad (2.27)$$

The pole contribution does not exhaust the full Schwarzschild response [13, 20, 22]. Schematically, the waveform may be decomposed as

$$\Psi(t, r_*) = \Psi_{\text{prompt}}(t, r_*) + \Psi_{\text{QNM}}(t, r_*) + \Psi_{\text{branch-cut}}(t, r_*), \quad (2.28)$$

where the prompt term describes the direct response generated when the perturbation first reaches the potential barrier, the QNM term is the discrete pole contribution derived above, and the branch-cut term is responsible for the late-time tail. These three pieces dominate different time regimes: the prompt response is relevant at early times, the pole contribution controls the intermediate ringdown stage, and the branch-cut contribution becomes important at sufficiently late times. In this work we focus on the ringdown window, where the discrete quasinormal contribution provides the natural description of the signal.

Within this regime, the pole expansion has a clear physical interpretation. The QNM frequencies ω_n are properties of the black-hole background and determine the oscillation and damping timescales of the response. The coefficients $\mathcal{C}_n$, however, are not universal: they depend on how the black hole is perturbed. The factorization

$$\mathcal{C}_n = B_n \mathcal{T}_n \quad (2.29)$$

separates these two ingredients. The excitation factor B_n depends only on the Schwarzschild background and on the normalization of the homogeneous solutions, whereas the overlap $\mathcal{T}_n$ contains the information about the initial data or source. This distinction is central for the rest of this work: Chapter 3 studies how controlled Gaussian initial data modify $\mathcal{T}_n$, while Chapter 5 compares the resulting theoretical amplitudes with time-domain numerical waveforms. This is the starting point of the spectral-filtering interpretation developed in Ref. [23].

Gaussian perturbations and spectral control

This chapter introduces the family of initial data used to probe quasinormal-mode excitation. We use localized Gaussian perturbations as smooth probes of the black-hole response, and add an oscillatory carrier to control the frequency content of the packet. The real-frequency spectrum gives useful intuition, but the QNM excitation is determined by the overlap evaluated at the complex frequencies ω_n . We first write the full radial overlap using Leaver’s representation of the Schwarzschild QNM eigenfunction, and then take its far-zone asymptotic limit, where the excitation mechanism becomes analytic. This construction follows the spectral-control framework of Ref. [23], adapted here to the notation and numerical conventions of this work.

3.1 Oscillatory Gaussian initial data

We consider source-free perturbations generated by the initial profile

$$f(r_*) \equiv \Psi(0, r_*) = A \exp\left[-\frac{(r_* - r_0)^2}{2\sigma^2}\right] \cos[\nu(r_* - r_0)], \quad (3.1)$$

where A is the amplitude, r_0 the initial position, σ the width, and ν the carrier frequency. The limit $\nu = 0$ gives a pure Gaussian.

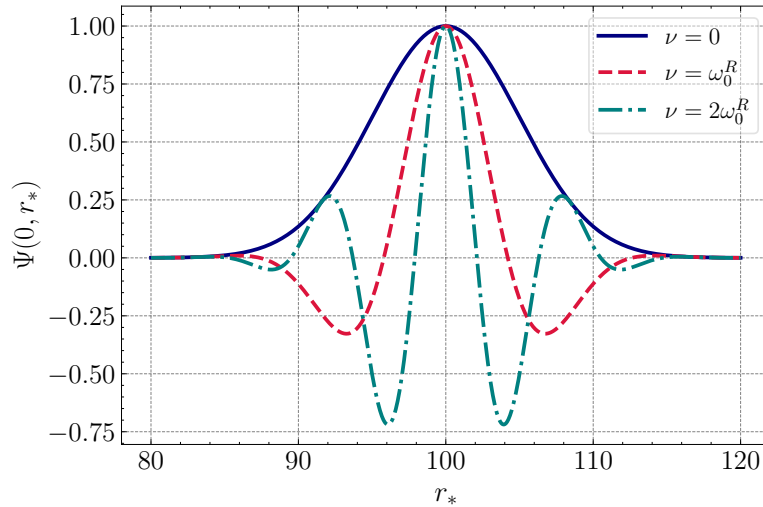


Figure 3.1: Oscillatory Gaussian initial data for fixed A , r_0 , and σ , and for different values of the carrier frequency ν . Increasing ν adds oscillatory structure to the packet while preserving the Gaussian envelope.

A pure Gaussian is a natural localized perturbation, but its real-frequency spectrum is centered at zero frequency. The role of the cosine is to shift spectral power toward finite frequencies, allowing us to test how the black hole responds when the initial data are tuned close to the oscillatory part of a QNM frequency.

The initial velocity is chosen to describe an ingoing perturbation in the asymptotic free-wave region,

$$g(r_*) \equiv \partial_t \Psi(0, r_*) = \partial_{r_*} f(r_*). \quad (3.2)$$

Indeed, when $V(r_*) \simeq 0$, the wave equation reduces to $\Psi(t, r_*) = F(t - r_*) + G(t + r_*)$. The choice $g = \partial_{r_*} f$ selects the component $G(t + r_*)$, which propagates toward decreasing r_* , i.e. toward the black-hole potential barrier.

3.2 Real- ω spectrum as intuition

The spatial Fourier transform of Eq. (3.1) is

$$\tilde{f}(\omega) = A\sqrt{\frac{\pi}{2}}\sigma e^{i\omega r_0} \left[e^{-\frac{\sigma^2}{2}(\omega-\nu)^2} + e^{-\frac{\sigma^2}{2}(\omega+\nu)^2} \right] \rightarrow |\tilde{f}(\omega)| \propto \exp\left[-\frac{\sigma^2}{2}(\omega^2 + \nu^2)\right] \cosh(\sigma^2\omega\nu). \quad (3.3)$$

For real ω , this expression displays two spectral branches centered near $\omega = \pm\nu$. The width σ controls their bandwidth, while the carrier frequency ν sets their location.

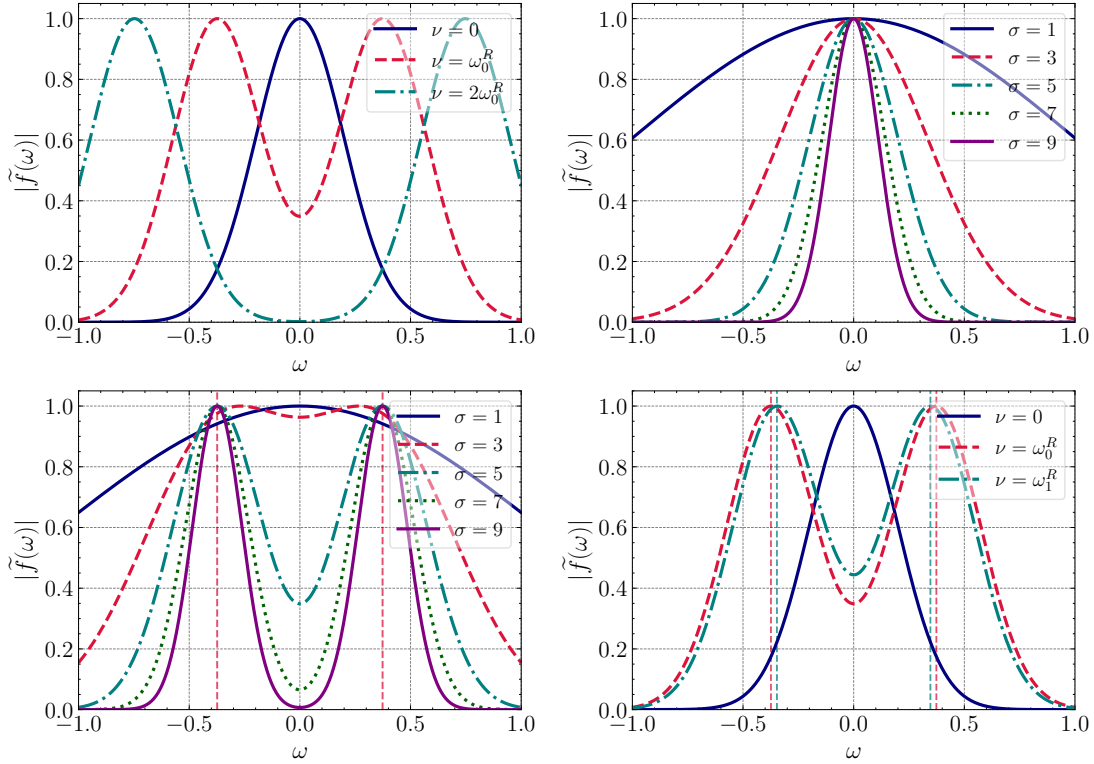


Figure 3.2: Real-frequency spectra of the oscillatory Gaussian initial data. Top left: varying ν at fixed σ shifts spectral power toward $\omega \simeq \pm\nu$. Top right: increasing σ narrows the pure-Gaussian spectrum around $\omega = 0$. Bottom left: for $\nu = \omega_0^R$, increasing σ sharpens the two spectral branches. Bottom right: Regge–Wheeler markers $\pm\omega_n^R$ indicate σ where the black hole samples the spectrum.

The role of σ and ν is already visible at the level of the real-axis Fourier spectrum.

Each spectral branch has width $\Delta\omega \sim \frac{1}{\sigma}$, while the branch separation is controlled by ν . The relevant dimensionless parameter is therefore

$$\alpha \equiv \sigma\nu.$$

For $\nu \neq 0$, using $y = \omega/\nu$, the curvature at the origin is $\frac{d^2}{dy^2} \log |\tilde{f}| \Big|_{y=0} = \alpha^2(\alpha^2 - 1)$. Hence $\alpha < 1$ implies that $\omega = 0$ is a maximum of the real-frequency spectrum, whereas for $\alpha > 1$ it becomes a

minimum and two maxima appear near $\omega \simeq \pm\nu$. The parameter α therefore separates an unresolved, low-frequency-centred spectrum from a frequency-selective one.

This real-axis picture is only a guide. It is useful for visualizing how the carrier frequency shifts the perturbation spectrum, but QNM excitation is controlled by the analytic continuation of the overlap to the complex frequencies ω_n .

3.3 Full radial overlap from Leaver's solution

For source-free initial data, the frequency-domain source term introduced in Chapter 2 reduces, using Eq. (3.2), to

$$I(\omega, r_*) = i\omega f(r_*) - g(r_*) = i\omega f(r_*) - \partial_{r_*} f(r_*). \quad (3.4)$$

In the wave-zone limit of the observer, the QNM pole contribution is controlled by the source overlap introduced in Eq. (2.24),

$$\mathcal{T}_n = \int_{-\infty}^{+\infty} \frac{\psi_{\text{in}}(r_*, \omega_n)}{A_{\text{out}}(\omega_n)} I(\omega_n, r_*) \, dr_*, \quad \mathcal{C}_n = B_n \mathcal{T}_n. \quad (3.5)$$

The factor B_n , defined in Eq. (2.23), depends only on the black-hole background, while $\mathcal{T}_n$ contains the dependence on the initial data.

The Leaver construction provides the full radial description of the Schwarzschild QNM eigenfunction entering this pole overlap [24]. More precisely, it gives a Frobenius–Jaffé series representation of the homogeneous radial solution satisfying the ingoing condition at the horizon and, through the minimal solution of the recurrence relation, the outgoing condition at infinity. It is therefore more complete than replacing the mode by its far-zone plane-wave behaviour.

We write

$$\frac{\psi_{\text{in}}(r_*, \omega_n)}{A_{\text{out}}(\omega_n)} = e^{i\omega_n r_*} \mathcal{W}_n^{\text{L}}(r_*), \quad \mathcal{W}_n^{\text{L}}(r_*) \rightarrow 1 \quad (r_* \rightarrow +\infty). \quad (3.6)$$

The corresponding full radial overlap is

$$\mathcal{T}_n^{\text{L}} = \int_{-\infty}^{+\infty} I(\omega_n, r_*) e^{i\omega_n r_*} \mathcal{W}_n^{\text{L}}(r_*) \, dr_*, \quad \mathcal{C}_n^{\text{L}} = B_n \mathcal{T}_n^{\text{L}}. \quad (3.7)$$

Thus Leaver's prescription keeps the same factorization $\mathcal{C}_n = B_n \mathcal{T}_n$, but retains the radial QNM profile inside the overlap integral.

To evaluate $\mathcal{W}_n^{\text{L}}$, we introduce $\rho \equiv \frac{r}{2M}$ and $z \equiv 1 - \frac{1}{\rho}$, so that the horizon is at $\rho = 1$ and spatial infinity corresponds to $z \rightarrow 1$. The reduced Leaver weight is represented as

$$\mathcal{W}_n^{\text{L}}(\rho) = z^{2p_n} \left(\sum_{j=0}^{N_{\text{max}}} a_j^{(n)} \right)^{-1} \sum_{j=0}^{N_{\text{max}}} a_j^{(n)} z^j, \quad p_n = -2iM\omega_n. \quad (3.8)$$

The normalization is chosen so that $\mathcal{W}_n^{\text{L}} \rightarrow 1$ at spatial infinity. The coefficients satisfy the three-term recurrence relation

$$\alpha_j a_{j+1}^{(n)} + \beta_j a_j^{(n)} + \gamma_j a_{j-1}^{(n)} = 0, \quad a_{-1}^{(n)} = 0, \quad a_0^{(n)} = 1, \quad (3.9)$$

with

$$\begin{aligned}\alpha_j &= j^2 + (2p_n + 2)j + (2p_n + 1), \\ \beta_j &= -\left[2(j + 2p_n)^2 + 2(j + 2p_n) + \ell(\ell + 1) - \epsilon\right], \\ \gamma_j &= j^2 + 4p_n j + 4p_n^2 - \epsilon - 1, \quad \epsilon = s^2 - 1.\end{aligned}\tag{3.10}$$

For gravitational perturbations $s = \pm 2$, hence $\epsilon = 3$. The minimal solution is obtained by backward Miller recursion and normalized with $a_0^{(n)} = 1$. In practice, the infinite series is truncated at finite $N_{\max}$. For the modes and radial domains considered here, $N_{\max} = 300$ gives stable values of $\mathcal{W}_n^L$ and of the corresponding overlap.

This construction should not be confused with a complete time-domain solution of the waveform. The Green-function response also contains the prompt part and the branch-cut contribution responsible for the late-time tail. Moreover, the single-integral form in Eq. (3.5) follows after taking the observer in the wave zone. The Leaver overlap is therefore the full radial version of the QNM pole overlap, not a replacement for the complete Green-function response.

3.4 Asymptotic limit and analytic overlap

The asymptotic approximation is obtained by taking the far-zone limit of the Leaver weight,

$$\mathcal{W}_n^L(r_*) \rightarrow 1, \quad r_* \rightarrow +\infty.\tag{3.11}$$

Equivalently, $\psi_{\text{in}}(r_*, \omega_n) \simeq A_{\text{out}}(\omega_n) e^{i\omega_n r_*}$. The overlap then reduces to

$$\mathcal{T}_n^{\text{asy}} = \int_{-\infty}^{+\infty} e^{i\omega_n r_*} I(\omega_n, r_*) dr_*.\tag{3.12}$$

Substituting Eq. (3.4) and integrating by parts, with vanishing boundary terms due to the Gaussian decay, gives

$$\mathcal{T}_n^{\text{asy}} = 2i\omega_n \int_{-\infty}^{+\infty} e^{i\omega_n r_*} f(r_*) dr_*.\tag{3.13}$$

Therefore, in the asymptotic approximation, QNM excitation samples the analytic continuation of the Fourier transform of the initial data at $\omega = \omega_n$.

Using Eq. (3.3), one obtains

$$\mathcal{T}_n^{\text{asy}} = iA\sqrt{2\pi} \sigma \omega_n e^{i\omega_n r_0} \left[e^{-\frac{\sigma^2}{2}(\omega_n - \nu)^2} + e^{-\frac{\sigma^2}{2}(\omega_n + \nu)^2} \right].\tag{3.14}$$

We use the convention $\omega_n = \omega_n^R - i\omega_n^I$, $\omega_n^I > 0$. The factor $e^{i\omega_n r_0}$ carries the translation of the Gaussian packet. For the asymptotic expression, this dependence factors out exactly. It is therefore natural to define

$$T_n^{\text{asy}} \equiv 2 \cdot e^{-i\omega_n r_0} \mathcal{T}_n^{\text{asy}}, \quad C_n^{\text{asy}} \equiv 2 \cdot e^{-i\omega_n r_0} \mathcal{C}_n^{\text{asy}} = B_n T_n^{\text{asy}}.\tag{3.15}$$

Explicitly,

$$T_n^{\text{asy}} = iA\sqrt{8\pi} \sigma \omega_n \left[e^{-\frac{\sigma^2}{2}(\omega_n - \nu)^2} + e^{-\frac{\sigma^2}{2}(\omega_n + \nu)^2} \right].\tag{3.16}$$

For the Leaver result we also use

$$C_n^L \equiv 2 \cdot e^{-i\omega_n r_0} \mathcal{C}_n^L.\tag{3.17}$$

This definition can be understood as referring the modal coefficient to the shifted retarded time $u - r_0 = t - r_*^{\text{obs}} - r_0$. It is only a time-origin convention and should not be interpreted as a sharp definition of the physical onset of ringdown. For the asymptotic overlap it also removes exactly the Gaussian translation factor. For the Leaver overlap, however, it does not remove the full r_0 -dependence. Thus shifting the Gaussian changes which part of the QNM radial wavefunction is sampled. Equation (3.16) is the main analytic result of the chapter. It gives a closed excitation landscape in (σ, ν) . The Leaver overlap is kept as the full radial validation of that far-zone prediction.

3.5 Excitation landscape

We now analyze the asymptotic expression and compare its predictions with the Leaver radial overlap. Taking the modulus of Eq. (3.16) gives

$$|T_n^{\text{asy}}|^2 = 16\pi A^2 \sigma^2 |\omega_n|^2 e^{-\sigma^2(P_n + \nu^2)} D_n(\sigma, \nu), \quad (3.18)$$

where

$$P_n = (\omega_n^R)^2 - (\omega_n^I)^2, \quad D_n = \cosh(2\sigma^2 \nu \omega_n^R) + \cos(2\sigma^2 \nu \omega_n^I). \quad (3.19)$$

For a fixed mode, multiplication by $|B_n|^2$ does not change the position of its extrema in (σ, ν) . For a pure Gaussian, $\nu = 0$, Eq. (3.18) reduces to

$$|T_n^{\text{asy}}|^2 \Big|_{\nu=0} = 32\pi A^2 \sigma^2 |\omega_n|^2 e^{-\sigma^2 P_n}. \quad (3.20)$$

If $P_n > 0$, equivalently $\omega_n^R > \omega_n^I$, the pure-Gaussian excitation has the finite optimal width

$$\sigma_n^* = \frac{1}{\sqrt{P_n}}. \quad (3.21)$$

For $P_n \leq 0$, this maximum is absent. Figure 3.3 shows the reduced coefficients $|C_n|$ for pure Gaussian data. The modes with $P_n > 0$, here $n = 0$ and $n = 1$, have finite maxima at $\sigma_n^* = 1/\sqrt{P_n}$. For the more strongly damped overtones, $P_n \leq 0$, this pure-Gaussian optimum is absent. The Leaver and asymptotic curves are nearly indistinguishable for this far-zone choice of r_0 .

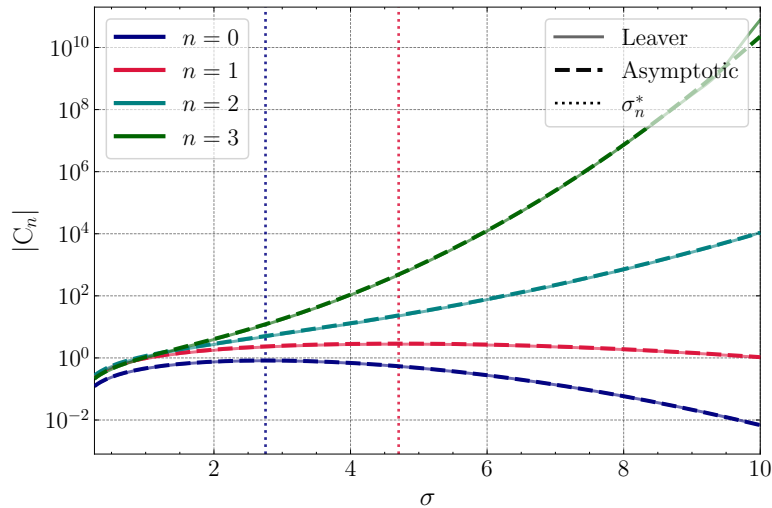


Figure 3.3: Pure-Gaussian excitation as a function of σ , for $\nu = 0$. The plotted quantity is the reduced modal coefficient $|C_n|$. Solid curves show the Leaver result and dashed curves the asymptotic prediction. Vertical dotted lines mark $\sigma_n^* = 1/\sqrt{P_n}$ when $P_n > 0$.

For $\nu \neq 0$, the extrema of Eq. (3.18) satisfy

$$\frac{\partial}{\partial \nu} \ln |T_n^{\text{asy}}|^2 = 0 \quad \Longleftrightarrow \quad \nu = \frac{S_n}{D_n}, \quad (3.22)$$

where

$$S_n = \omega_n^R \sinh(2\sigma^2 \nu \omega_n^R) - \omega_n^I \sin(2\sigma^2 \nu \omega_n^I). \quad (3.23)$$

In the frequency-selective regime, the positive-frequency branch dominates,

$$e^{2\sigma^2 \nu \omega_n^R} \gg 1,$$

and the stationarity condition reduces to

$$\nu \simeq \omega_n^R \tanh(2\sigma^2 \nu \omega_n^R) \simeq \omega_n^R. \quad (3.24)$$

This is the formal origin of the spectral-targeting rule: the carrier frequency should be placed near the real oscillation frequency of the mode. The actual excitation, however, is controlled by the full complex-frequency expression.

The transition away from the pure-Gaussian maximum is seen from

$$\left. \frac{\partial^2}{\partial \nu^2} \ln |T_n^{\text{asy}}|^2 \right|_{\nu=0} = 2\sigma^2(\sigma^2 P_n - 1). \quad (3.25)$$

For $P_n > 0$, $\nu = 0$ changes from a local maximum to a local minimum when σ crosses σ_n^* . Thus sufficiently wide Gaussians become frequency-selective: the excitation is no longer maximized by a pure Gaussian, but by nonzero carrier frequencies.

Figure 3.4 shows this behaviour for $\sigma = 5M$. Each mode is normalized by its own maximum to emphasize the location of the resonance. The weakly damped modes peak close to $\nu = \omega_n^R$, while the more damped mode displays a broader and shifted maximum. This illustrates why the real-axis rule $\nu \simeq \omega_n^R$ is useful but not exact: the overlap is evaluated at the full complex frequency ω_n .

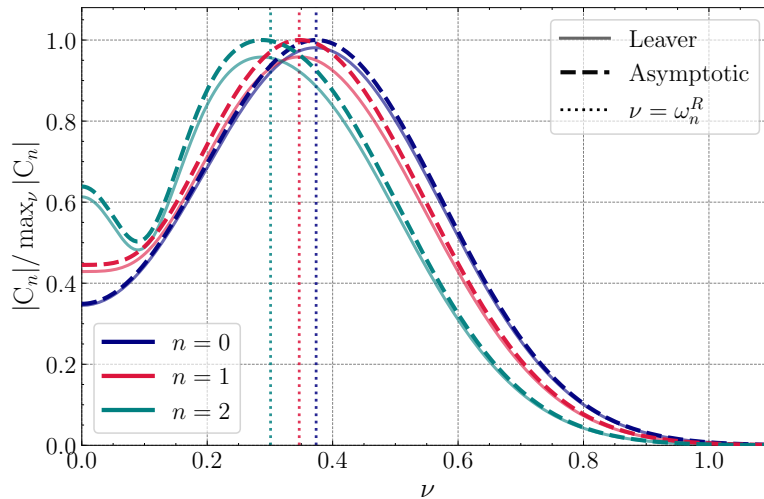


Figure 3.4: Reduced QNM excitation coefficients $|C_n|$ as a function of the carrier frequency ν , for fixed $\sigma = 5M$ and $\ell = 2$. Each mode is normalized by its maximum over ν . Solid curves show the Leaver result and dashed curves the asymptotic prediction. Vertical dotted lines indicate $\nu = \omega_n^R$.

For the fundamental Regge–Wheeler mode with $\ell = 2$ and $M = 1$, the numerical values used are computed with the `qnm` package [25] and consistent with standard Schwarzschild tabulations. The excitation factors B_n used in Figs. 3.3 and 3.4 are taken from Berti’s ringdown data in the Schwarzschild limit [26, 27].

For reference, Table 3.1 lists the Schwarzschild QNM frequencies used in this work, together with derived quantities that appear repeatedly in the analytic estimates. All values are given in units $M = 1$.

Mode (ℓ, n)	ω_n^R	ω_n^I	σ_n^*
(2, 0)	0.373672	0.088962	2.755
(2, 1)	0.346711	0.273915	4.705
(2, 2)	0.301053	0.478277	–
(2, 3)	0.251505	0.705148	–

Table 3.1: Reference Schwarzschild Regge–Wheeler QNM frequencies for the $\ell = 2$ gravitational sector. We use the convention $\omega_n = \omega_n^R - i\omega_n^I$, with $\omega_n^I > 0$. The quantity σ_n^* is shown only when $P_n > 0$. The frequencies are computed with the `qnm` package and agree with standard Schwarzschild tabulations [13, 25, 27].

3.6 Interpretation

The oscillatory Gaussian family provides a controlled way of probing black-hole resonances. The width σ controls the bandwidth, the carrier frequency ν shifts the spectral content, and the overlap T_n measures how efficiently the initial data excite each mode.

The Leaver overlap gives the full radial version of the QNM pole prediction. It keeps the factorization $C_n = B_n \mathcal{T}_n$, but retains the complete radial QNM profile through $\mathcal{W}_n^L$. This makes it suitable for testing the excitation mechanism beyond the far-zone approximation. However, because the radial profile is represented by a series and remains inside the integral, it does not lead to a simple closed-form landscape in (σ, ν) .

The asymptotic limit $\mathcal{W}_n^L \rightarrow 1$ removes this radial weight and gives the analytic expression (3.16). In that limit, the black hole acts as a resonant spectral filter: modes are enhanced when the analytically continued perturbation has significant support near ω_n , and suppressed otherwise [23]. The pure-Gaussian case shows how the damping of the mode controls whether a finite optimal width exists, while the oscillatory family shows how the carrier frequency can selectively target modes through $\nu \simeq \omega_n^R$.

The reduced coefficients C_n are useful for comparing the intrinsic dependence on σ and ν . In the asymptotic expression, removing $e^{i\omega_n r_0}$ exactly factors out the initial position of the packet. In the Leaver calculation, the same definition fixes a common time-origin convention but does not remove the residual r_0 -dependence, which remains encoded in the radial weight $\mathcal{W}_n^L$.

Thus, the asymptotic result gives the clean analytic mechanism, while the Leaver overlap tests how much of that mechanism survives when the full Schwarzschild radial structure is included. This distinction will be used in Chapter 5, where these predictions are compared with time-domain numerical evolutions.

Numerical time-domain framework

This chapter describes the numerical framework used to generate the time-domain waveforms analyzed in the following chapters. The method is formulated for a generic one-dimensional wave equation with an effective potential and the evolution uses fourth-order finite differences in space and a fourth-order Runge–Kutta scheme in time. The explicit finite-difference stencils, including the one-sided boundary closures, are collected in Appendix D.

4.1 Evolution equation

We consider the wave equation

$$\partial_t^2 \Psi - \partial_{r_*}^2 \Psi + V(r_*) \Psi = 0, \quad (4.1)$$

where r_* denotes the spatial coordinate used in the numerical evolution. For Schwarzschild perturbations, the Regge–Wheeler potential is naturally written as a function of the Schwarzschild radius r , while the evolution is performed in the tortoise coordinate r_* . It is therefore evaluated in the code as $V_\ell^{RW}(r(r_*))$, where $r(r_*) = 2M \left[1 + W_0 \left(e^{r_*/(2M)-1} \right) \right]$, and W_0 is the principal branch of the Lambert W function.

Introducing

$$u(t, r_*) \equiv \Psi(t, r_*), \quad v(t, r_*) \equiv \partial_t \Psi(t, r_*), \quad (4.2)$$

Eq. (4.1) becomes

$$\partial_t u = v, \quad \partial_t v = L[u], \quad L[u] \equiv \partial_{r_*}^2 u - V(r_*)u. \quad (4.3)$$

This first-order system is the one evolved numerically.

4.2 Grid and time integration

The spatial domain is discretized on a uniform grid,

$$r_{*,j} = r_{*,\min} + jh, \quad j = 0, \dots, N-1, \quad h = \frac{r_{*,\max} - r_{*,\min}}{N-1}. \quad (4.4)$$

The time step is chosen according to a Courant–Friedrichs–Lewy (CFL) factor,

$$\Delta t = \text{CFL} \cdot h. \quad (4.5)$$

The same Courant factor is used in all resolution comparisons; throughout this work we set $\text{CFL} = 0.1$.

The spatial operator is discretized as

$$L_h[u]_j = D_h^{(2)} u_j - V_j u_j, \quad V_j \equiv V(r_{*,j}), \quad (4.6)$$

where $D_h^{(2)}$ denotes the fourth-order finite-difference approximation to $\partial_{r_*}^2$. In the bulk this is a centered five-point stencil, while near the boundaries one-sided fourth-order closures are used. The corresponding first- and second-derivative formulas are listed in Appendix D.

Writing

$$Y = \begin{pmatrix} u \\ v \end{pmatrix}, \quad F(Y) = \begin{pmatrix} v \\ L_h[u] \end{pmatrix}, \quad (4.7)$$

one step of the fourth-order Runge–Kutta method is

$$\begin{aligned} K_1 &= F(Y^n), & K_2 &= F\left(Y^n + \frac{\Delta t}{2} K_1\right), \\ K_3 &= F\left(Y^n + \frac{\Delta t}{2} K_2\right), & K_4 &= F(Y^n + \Delta t K_3), \end{aligned} \quad (4.8)$$

followed by

$$Y^{n+1} = Y^n + \frac{\Delta t}{6} (K_1 + 2K_2 + 2K_3 + K_4). \quad (4.9)$$

Thus both the spatial discretization and the time integrator are fourth-order accurate.

4.3 Initial data and boundary treatment

The initial profile is the Gaussian packet introduced in Eq. (3.1). To generate an initially ingoing packet in the asymptotic free-wave region, the initial velocity is chosen as

$$v(0, r_*) = +\partial_{r_*} u(0, r_*). \quad (4.10)$$

Indeed, when $V(r_*) \simeq 0$, the wave equation reduces locally to the free equation, whose general solution is

$$u(t, r_*) = F(t - r_*) + G(t + r_*). \quad (4.11)$$

The choice $v = \partial_{r_*} u$ selects the $G(t + r_*)$ component, corresponding to propagation toward decreasing r_* , i.e. toward the potential barrier from the right. The derivative in Eq. (4.10) is evaluated with the fourth-order first-derivative operator listed in Appendix D.

Since the computational domain is finite, outgoing Sommerfeld conditions are imposed at both ends. In the regions where $V(r_*) \simeq 0$, outgoing waves satisfy

$$v = +\partial_{r_*} u \quad \text{at } r_* = r_{*,\min}, \quad v = -\partial_{r_*} u \quad \text{at } r_* = r_{*,\max}. \quad (4.12)$$

These conditions are reimposed at every Runge–Kutta substep using the one-sided fourth-order first-derivative stencils of Appendix D.

To reduce possible boundary contamination, the numerical potential can be windowed as

$$V_{num}(r_*) = W(r_*)V(r_*), \quad W(r_*) \simeq 1 \text{ in the physical region}, \quad (4.13)$$

with $W(r_*)$ smoothly suppressing the potential near the numerical edges. Although it has not been activated in this work, an optional sponge layer can also be used to damp outgoing radiation close to the boundaries.

Since r_*^{obs} does not necessarily coincide with a grid point, the value of the field is reconstructed by local fourth-order interpolation.

4.4 Fourth-order convergence

The code is validated through convergence tests. In the free case, $V = 0$, the numerical waveform can be compared with the exact free-wave solution. If the scheme is fourth-order accurate, the error satisfies

$$\Psi_h - \Psi_{exact} = \mathcal{O}(h^4). \quad (4.14)$$

Equivalently, for three successive resolutions h , $h/2$, and $h/4$, the self-convergence ratio

$$Q_h(t) = \frac{\Psi_h(t) - \Psi_{h/2}(t)}{\Psi_{h/2}(t) - \Psi_{h/4}(t)} \quad (4.15)$$

should approach

$$Q_h \simeq 2^4 = 16 \quad (4.16)$$

in the resolved part of the signal, away from zeros of the denominator. A global estimate of the convergence order is obtained from

$$p = \log_2 \left(\frac{\|\Psi_h - \Psi_{h/2}\|}{\|\Psi_{h/2} - \Psi_{h/4}\|} \right), \quad (4.17)$$

which should tend to $p = 4$.

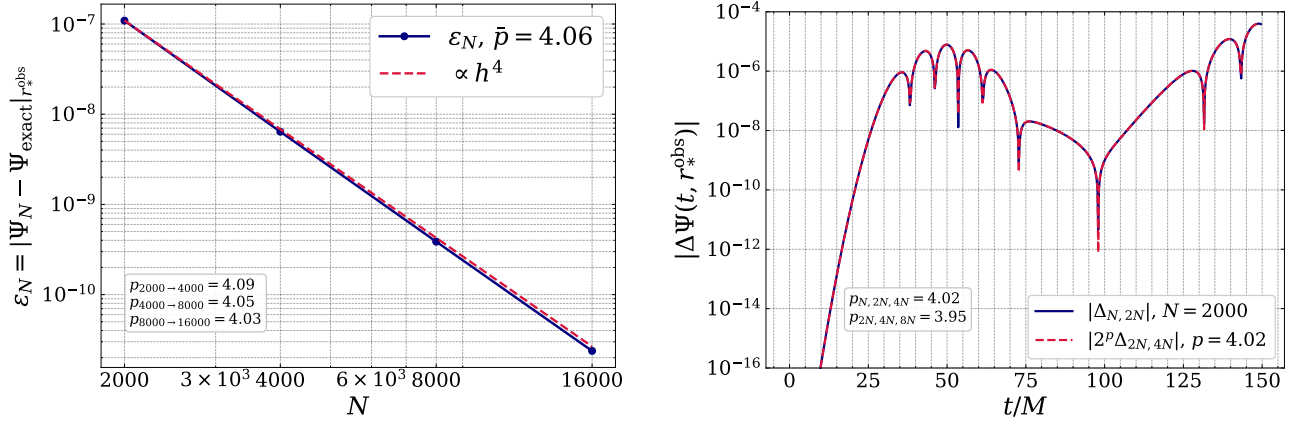


Figure 4.1: **Fourth-order convergence tests.** Left: direct convergence test in the free case, $V = 0$, where the numerical waveform is compared with the exact solution. Right: self-convergence test for an evolution with an effective potential. Both diagnostics confirm fourth-order convergence in the resolved part of the waveform.

These tests validate the numerical waveforms used as benchmarks for the quasinormal-mode reconstructions developed in the following chapters.

Schwarzschild ringdown and resonances

The previous chapters introduced the Green-function framework, the factorization $\mathcal{C}_n = B_n \mathcal{T}_n$, and the oscillatory Gaussian initial data used to control the spectral content of the perturbation. We now combine these ingredients with the time-domain Regge–Wheeler evolution of Chapter 4. The goal is to study how the Schwarzschild waveform, the fundamental QNM excitation, and the late-time tail depend on the Gaussian parameters σ and ν , thereby testing the spectral-filtering picture of Ref. [23].

Throughout this chapter we use geometrized units with $M = 1$, focus on the dominant gravitational multipole $\ell = 2$, and set $A = 1$ unless otherwise stated. The numerical evolutions are performed on $r_* \in [-500M, 1500M]$ with $N = 10000$, CFL = 0.1, and $T_F = 1400M$.

5.1 Waveform dependence on the spectral parameters (σ, ν)

We now evolve axial perturbations of a Schwarzschild black hole using the time-domain framework of Chapter 4. The initial data are the oscillatory Gaussian profiles introduced in Eq. (3.1), with the ingoing prescription of Eq. (3.2).

Figure 5.1 shows the effect of the two parameters that control the initial packet. As discussed in Sec. 3.2, the width σ controls the spectral bandwidth, while the carrier frequency ν shifts the spectral content away from $\omega = 0$. The dimensionless combination $\alpha = \sigma\nu$ measures whether the oscillatory modulation is resolved inside the Gaussian envelope: for small α the packet is still mainly low-frequency, whereas for larger α the spectrum is displaced toward frequencies of order $\pm\nu$. This is only a real-frequency diagnostic; the actual QNM amplitudes are determined by the complex-frequency overlaps of Chapter 3.

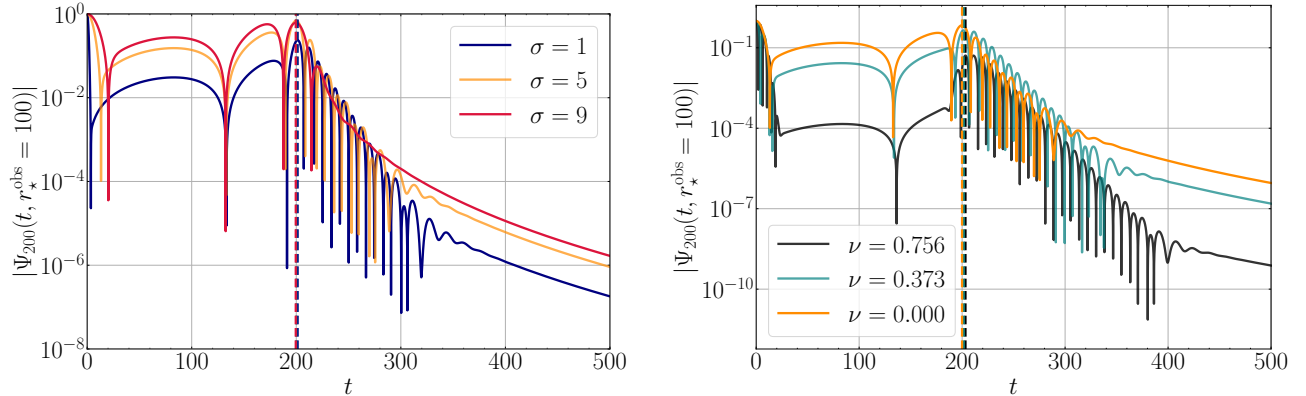


Figure 5.1: Numerical RW waveforms extracted at $r_*^{obs} = 100M$, with initial packets centred at $r_0 = 100M$. Left: pure Gaussian data, $\nu = 0$, for different widths σ . Right: oscillatory Gaussian data with fixed $\sigma = 5M$ and different carrier frequencies ν . The dashed vertical line marks the approximate free-propagation time $t \simeq r_0 + r_*^{obs}$, used only as a visual reference. Adapted from Ref. [23].

The left panel corresponds to pure Gaussian data, $\nu = 0$, for which $\alpha = 0$ for any value of σ . Increasing σ narrows the spectrum but keeps it centred at $\omega = 0$, so it does not move the initial data toward the fundamental QNM frequency. Wider pure Gaussians may therefore enhance the relative low-frequency contribution rather than the QNM amplitude itself. In the Green-function picture, this is expected: the Schwarzschild response contains both QNM poles, which generate the damped oscillatory ringdown, and a branch-cut contribution, responsible for the late-time power-law tail [20, 28, 29]. Thus a larger late-time signal should not automatically be interpreted as a stronger excitation of C_0 .

The right panel isolates the effect of ν at fixed $\sigma = 5M$. As ν increases, the initial spectrum is shifted away from $\omega = 0$. This suppresses the low-frequency part of the response and can make the oscillatory regime persist for longer.

For the fundamental $\ell = 2$ mode, Table 3.1 gives $\omega_0^R \simeq 0.373/M$. Therefore the waveform with $\nu \simeq 0.373/M$ has the largest fundamental-mode amplitude among the examples shown, because it is the least suppressed in the overlap. By contrast, $\nu \simeq 2\omega_0^R$ still reduces the low-frequency content and can extend the visible oscillatory part, but it is displaced from the fundamental frequency and therefore gives a smaller C_0 .

Thus the waveform dependence on (σ, ν) involves two related but different effects. Moving spectral weight away from $\omega = 0$ reduces the relative importance of the tail and can lengthen the visible oscillatory regime. Placing that spectral weight near ω_0^R reduces the exponential suppression of the fundamental overlap and increases the amplitude of C_0 . The following sections make this interpretation quantitative by comparing the numerical waveforms with theoretical reconstructions of Chapter 3.

5.2 Asymptotic and Leaver QNM reconstructions

We now compare the numerical waveform with the QNM pole reconstructions predicted by the asymptotic and Leaver overlaps of Chapter 3. The QNM frequencies are computed with `qnm` [25], and the excitation factors B_n are taken from Berti’s tabulated values in the Schwarzschild limit [26, 27]. The two theoretical modal coefficients are $\mathcal{C}_n^{\text{asy}} = B_n \mathcal{T}_n^{\text{asy}}$ and $\mathcal{C}_n^{\text{L}} = B_n \mathcal{T}_n^{\text{L}}$. The corresponding real waveforms are then constructed using the pole-pair prescription of Eq. (2.27).

Figure 5.2 compares the numerical waveform with the two theoretical reconstructions over the QNM-dominated part of the signal. Both reconstructions capture the ringdown with good accuracy after the initial prompt response. In this far-zone configuration, the asymptotic prediction already gives a very accurate description of the envelope. This confirms that Leaver does not introduce a different excitation mechanism, but only replaces the far-zone kernel by the full radial QNM profile. The dependence on r_0 , where the Leaver correction becomes more relevant as the packet approaches the RW barrier, is analysed below.

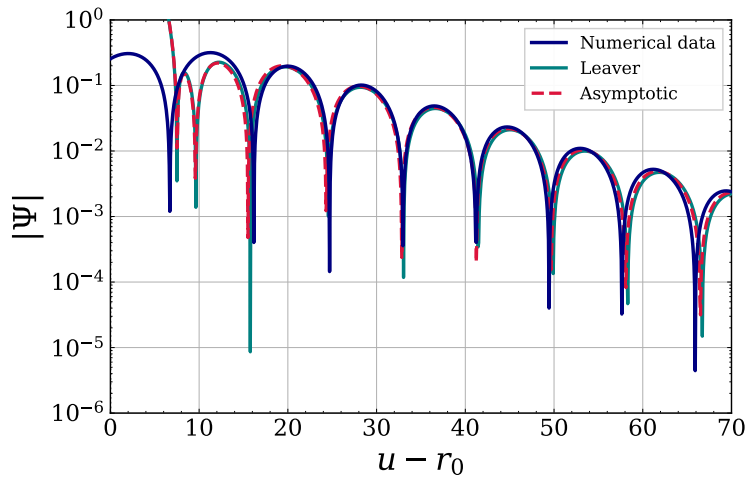


Figure 5.2: **Asymptotic and Leaver QNM reconstructions.** Comparison between the numerical waveform and the QNM reconstructions obtained from $\mathcal{T}_n^{\text{asy}}$ and $\mathcal{T}_n^{\text{L}}$. The simulation uses Schwarzschild Regge–Wheeler perturbations with $r_0 = 100M$, $\sigma = 5M$, $\nu = \omega_0^R$, and $r_*^{\text{obs}} = 100M$.

5.3 Time-domain validation of the excitation landscape

We now test the excitation picture directly in the time domain. The numerical waveforms are evaluated at fixed observer radius using the retarded time $u = t - r_*^{obs}$, and the QNM amplitudes are referred to a common time origin,

$$\tau = u - u_{ref}. \quad (5.1)$$

For scans at fixed initial position we take $u_{ref} = r_0$. When r_0 is varied, this convention is applied separately to each waveform, so that the reported coefficient is always measured relative to the corresponding initial pulse position.

In each sliding window starting at u_0 , the waveform is fitted with fixed QNM frequencies. For the fundamental mode the fitting model is

$$\Psi(u) \simeq \text{Re} \left[C_0^{num}(u_0) e^{-i\omega_0(u-u_{ref})} \right]. \quad (5.2)$$

The coefficient used in the scans is obtained by coherently averaging the complex local estimates over the selected ringdown interval I_{RD} ,

$$|C_0^{num}| = \left| \langle C_0^{num}(u_0) \rangle_{u_0 \in I_{RD}} \right|. \quad (5.3)$$

In addition, to compare the numerical and theoretical amplitudes at the same time reference, we write the theoretical QNM contribution as

$$\Psi_{QNM}(u) = -\text{Re} \left[C_0 e^{-i\omega_0(u-r_0)} \right], \quad C_0 = B_0 T_0. \quad (5.4)$$

The shaded bands in the following scans quantify the sensitivity of the result to the choice of the extraction window. They are obtained by repeating the coherent average over a family of admissible ringdown intervals. Therefore, they should be interpreted as window-sensitivity bands, not as statistical confidence intervals. The full extraction prescription is given in Appendix C.

Figure 5.3 illustrates the procedure for a representative waveform. The fundamental amplitude develops a stable intermediate region, while the higher overtones are more sensitive to the fitting window. This motivates using the coherently averaged fundamental coefficient as the main diagnostic in the parameter scans.

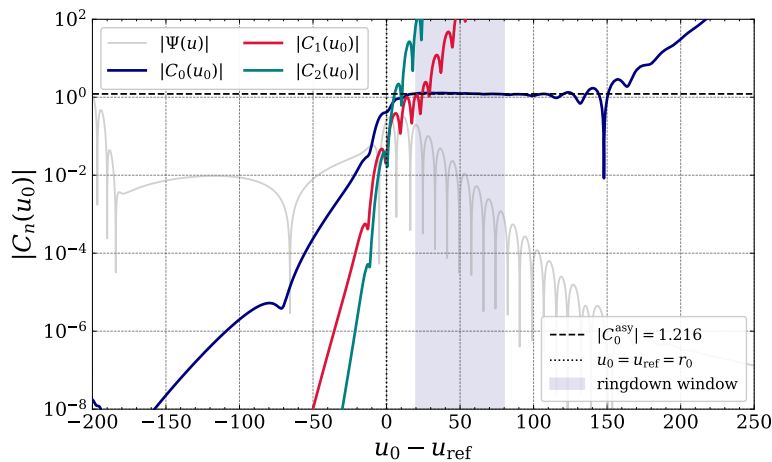


Figure 5.3: Sliding-window extraction of local QNM amplitudes for a representative Regge–Wheeler waveform with $r_0 = 100M$, $r_*^{obs} = 100M$, $\sigma = 5M$, and $\nu = 0.47M^{-1}$. The gray curve shows $|\Psi(u)|$, the colored curves show the local amplitudes $|C_n(u_0)|$, and the shaded region marks the ringdown interval used for the coherent extraction of $|C_0^{num}|$.

5.3.1 Dependence on the spectral parameters

We now apply this protocol to scans over the carrier frequency ν , the width σ , and the initial-data position r_0 . In each case the numerically extracted coefficient is compared with the Leaver and asymptotic predictions after all amplitudes have been expressed in the same reference-time convention.

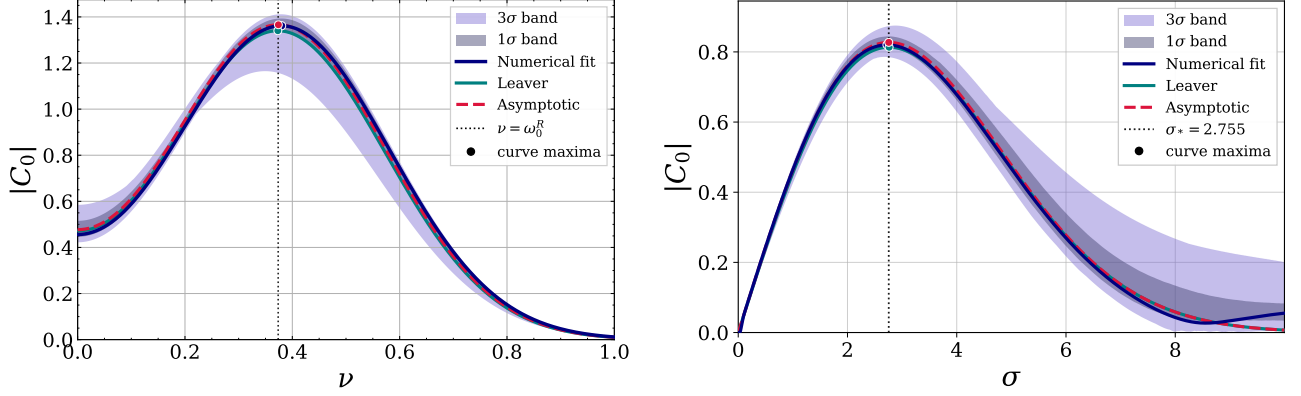


Figure 5.4: Time-domain validation of the fundamental excitation landscape. Left: $|C_0|$ as a function of the carrier frequency ν , for fixed $\sigma = 5M$. Right: $|C_0|$ as a function of the Gaussian width σ , for fixed $\nu = 0$. Both panels use $r_0 = 100M$, $r_*^{obs} = 100M$, and the common reference $u_{ref} = r_0$. The numerical curves are obtained from the coherent time-domain extraction of C_0 , while the Leaver and asymptotic curves are evaluated in the same common time reference. The shaded regions show the window-to-window dispersion of the numerical extraction. In the ν -scan, the vertical marker denotes $\nu = \omega_0^R$. In the σ -scan, the vertical marker denotes σ_* .

The main numerical values extracted from the scans are summarized in Table 5.1. The carrier-frequency scan confirms the expected resonant behaviour: the excitation is maximized close to $\nu = \omega_0^R$. The numerical maximum is slightly shifted with respect to the theoretical curves, but the agreement is at the percent level in amplitude and very close in peak location.

Scan	Numerical extraction	Leaver/asymptotic curves	Analytic estimate
ν -scan, $\sigma = 5$	$\nu_{\max}^{\text{num}} = 0.380$	$\nu_{\max}^L = 0.373$, $\nu_{\max}^{\text{asy}} = 0.374$	$\nu \simeq \omega_0^R = 0.373672\dots$
	$ C_0 _{\max}^{\text{num}} = 1.362$	$ C_0 _{\max}^L = 1.340$, $ C_0 _{\max}^{\text{asy}} = 1.366$	$ C_0^{\text{asy}}(\nu = \omega_0^R) = 1.36598$
σ -scan, $\nu = 0$	$\sigma_{\max}^{\text{num}} = 2.711$	$\sigma_{\max}^L = 2.762$, $\sigma_{\max}^{\text{asy}} = 2.755$	$\sigma_* = 2.7554\dots$
	$ C_0 _{\max}^{\text{num}} = 0.826$	$ C_0 _{\max}^L = 0.813$, $ C_0 _{\max}^{\text{asy}} = 0.827$	$ C_0^{\text{asy}}(\sigma = \sigma_*) = 0.82719$

Table 5.1: Summary of the peak locations and amplitudes for the scans shown in Fig. 5.4. The numerical column gives the direct time-domain extraction. The Leaver/asymptotic column gives the maxima found from the corresponding theoretical curves evaluated numerically in the same common reference. The last column displays the analytic estimates.

The agreement between the numerical peak, the Leaver prediction and the asymptotic curve therefore provides a direct time-domain validation of the resonance condition $\nu = \frac{S_0}{D_0} \simeq \omega_0^R$. This is precisely the mechanism suggested by the real-frequency picture of the initial data, but the actual excitation is controlled by the overlap evaluated at the complex QNM frequency.

The width scan probes a complementary effect. For $\nu = 0$, the excitation is not maximized by making the Gaussian arbitrarily narrow or arbitrarily broad. Instead, the asymptotic overlap predicts

a finite optimal width, $\sigma_* = [(\omega_0^R)^2 - (\omega_0^I)^2]^{-1/2} = 2.7554$, in close agreement with the numerical maximum $\sigma_{\max}^{\text{num}} \simeq 2.711$.

5.3.2 Dependence on the initial-data position

We finally test the dependence of the extracted fundamental amplitude on the initial position r_0 . We use the same time reference convention used in Eq. (5.4)

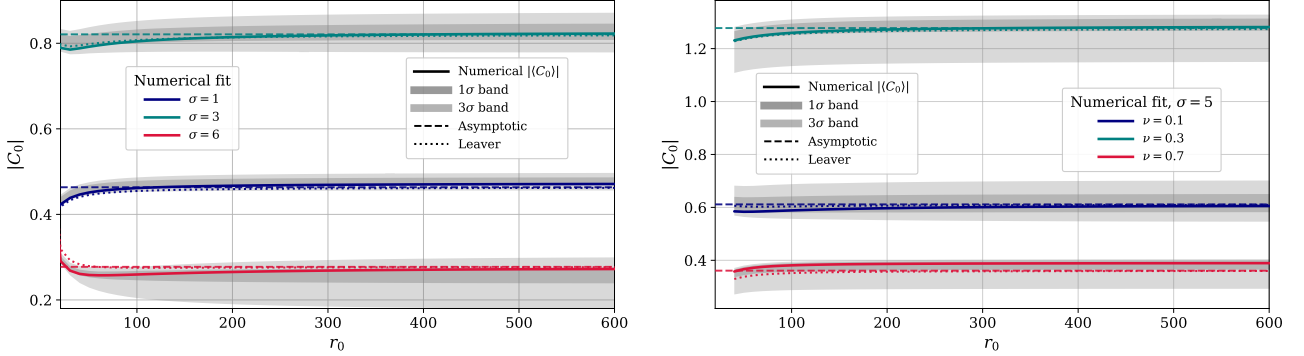


Figure 5.5: Dependence of the coherently extracted fundamental amplitude on the initial-data position r_0 . Left: pure Gaussian data, $\nu = 0$, for different widths σ . Right: fixed width $\sigma = 5M$, for different carrier frequencies ν . The shaded regions show the extraction-window sensitivity bands.

The r_0 -scans show a clear convergence between the time-domain extraction and the frequency-domain predictions as the initial perturbation is moved farther away from the potential barrier. This agreement is observed for the pure-Gaussian scan and for most carrier frequencies at fixed $\sigma = 5M$. The main exception is the largest value shown, $\nu = 0.7M^{-1}$, for which the numerical amplitude remains systematically higher over part of the scan, although still within the extraction-window bands.

5.4 Late-time tail suppression

The previous sections focused on the pole contribution to the Green-function response and on the excitation of the fundamental QNM. The Schwarzschild response, however, is not exhausted by the discrete pole sum. As discussed in Eq. (2.28), the inverse Green-function contour also contains a branch-cut contribution, responsible for the late-time tail [20, 22, 30]. In the timelike regime at fixed radius, this contribution gives Price’s power-law decay, $\Psi_\ell^{\text{tail}} \sim t^{-(2\ell+3)}$ [28, 31]. In this section we use a more detailed version of this statement, following Ref. [32]: Leaver’s far-zone late-time expansion, which makes explicit how the tail amplitude depends on the Cauchy data.

Let $x \equiv r_*$, $\psi_0(x) \equiv \Psi(0, x)$, and $v_0(x) \equiv \partial_t \Psi(0, x)$. We define the Leaver moments as $I_\ell[q] = \int_{-\infty}^{+\infty} w_\ell(x) q(x) dx$, with $w_\ell(x) \equiv r(x)^{\ell+1}$. Then, the leading far-zone branch-cut contribution is

$$\Psi_\ell^{\text{tail}}(t, x_{\text{obs}}) \sim \frac{4(-1)^\ell}{[(2\ell+1)!!]^2} r_{\text{obs}}^{\ell+1} \left[(2\ell+3)! I_\ell[\psi_0] (t-t_0)^{-(2\ell+4)} - (2\ell+2)! I_\ell[v_0] (t-t_0)^{-(2\ell+3)} \right], \quad (5.5)$$

where $r_{\text{obs}} = r(x_{\text{obs}})$. This expression is a very-late-time, far-zone expansion of the branch-cut part of the response. The important point for the present analysis is that the tail amplitude is not set only by the Green function, but also by the projection of the initial data onto the Leaver moments $I_\ell[\psi_0]$ and

$I_\ell[v_0]$. For the dominant gravitational multipole used in the simulations, $\ell = 2$, Eq. (5.5) becomes

$$\Psi_2^{\text{tail}}(t, x_{\text{obs}}) \simeq a_7 (t - t_0)^{-7} + a_8 (t - t_0)^{-8}, \quad (5.6)$$

with $a_7 = -\frac{4(6)!}{(5!!)^2} r_{\text{obs}}^3 I_2[v_0]$, and $a_8 = \frac{4(7)!}{(5!!)^2} r_{\text{obs}}^3 I_2[\psi_0]$. Thus the leading t^{-7} coefficient is controlled by $I_2[v_0]$, while the subleading t^{-8} coefficient is controlled by $I_2[\psi_0]$. Using the ingoing Gaussian data of Eqs. (3.1)–(3.2)

$$I_\ell[v_0] = \int_{-\infty}^{+\infty} w_\ell(x) \partial_x \psi_0(x) dx = - \int_{-\infty}^{+\infty} w'_\ell(x) \psi_0(x) dx, \quad (5.7)$$

where the boundary term vanishes because the packet is localized.

We now set $x - r_0 = \sigma u$, and $\alpha \equiv \sigma \nu$, so that the Gaussian profile becomes $\psi_0(u) = A e^{-u^2/2} \cos(\alpha u)$. The relevant integrals are

$$J_n(\alpha) \equiv \int_{-\infty}^{+\infty} u^n e^{-u^2/2} \cos(\alpha u) du = \sqrt{2\pi} e^{-\alpha^2/2} \text{Re} [(-i)^n \text{He}_n(\alpha)], \quad (5.8)$$

where He_n denotes the probabilists' Hermite polynomial. This follows from the Gaussian Fourier transform $\int_{-\infty}^{+\infty} e^{-u^2/2} e^{i\alpha u} du = \sqrt{2\pi} e^{-\alpha^2/2}$. Expanding the radial weights around the centre of the packet, $w_\ell(r_0 + \sigma u) = \sum_{n=0}^{\infty} \frac{(\sigma u)^n}{n!} w_\ell^{(n)}(r_0)$, and equivalently for $w'_\ell(r_0 + \sigma u)$, one obtains

$$I_\ell[v_0] = -A\sigma \sum_{n=0}^{\infty} \frac{\sigma^n}{n!} w_\ell^{(n+1)}(r_0) J_n(\alpha), \quad I_\ell[\psi_0] = A\sigma \sum_{n=0}^{\infty} \frac{\sigma^n}{n!} w_\ell^{(n)}(r_0) J_n(\alpha). \quad (5.9)$$

Only even n contribute, because $J_n = 0$ for odd n . Defining the dimensionless derivatives $b_j^\ell \equiv r_0^{j-\ell-1} \left. \frac{d^j w_\ell}{dx^j} \right|_{x=r_0}$, the common exponential factor can be extracted explicitly:

$$I_\ell[v_0] = -A\sqrt{2\pi} \sigma r_0^\ell e^{-\alpha^2/2} \mathcal{B}_v^\ell(\alpha; r_0, \sigma), \quad I_\ell[\psi_0] = A\sqrt{2\pi} \sigma r_0^{\ell+1} e^{-\alpha^2/2} \mathcal{B}_\psi^\ell(\alpha; r_0, \sigma) \quad (5.10)$$

where

$$\mathcal{B}_v^\ell(\alpha; r_0, \sigma) = \sum_{k=0}^{\infty} \left(\frac{\sigma}{r_0} \right)^{2k} \frac{b_{2k+1}^\ell}{(2k)!} (-1)^k \text{He}_{2k}(\alpha), \quad \mathcal{B}_\psi^\ell(\alpha; r_0, \sigma) = \sum_{k=0}^{\infty} \left(\frac{\sigma}{r_0} \right)^{2k} \frac{b_{2k}^\ell}{(2k)!} (-1)^k \text{He}_{2k}(\alpha). \quad (5.11)$$

This is the useful factorization: the dependence on the oscillatory structure appears through the universal Gaussian factor $e^{-\alpha^2/2}$, while the radial corrections are organized in powers of $(\sigma/r_0)^{2k}$.

In the far-field limit $2M/r_0 \ll 1$, the tortoise and Schwarzschild radial coordinates coincide to leading order, $r(x)^{\ell+1} \sim x^{\ell+1}$. Keeping the leading terms of the moments for $\ell = 2$ gives

$$a_7 \simeq \frac{192}{5} A\sqrt{2\pi} \sigma r_{\text{obs}}^3 r_0^2 e^{-\alpha^2/2}, \quad a_8 \simeq \frac{448}{5} A\sqrt{2\pi} \sigma r_{\text{obs}}^3 r_0^3 e^{-\alpha^2/2}. \quad (5.12)$$

Equations (5.10)–(5.12) show that both Leaver moments, and therefore both tail coefficients, contain the same dominant suppression factor, $a_7, a_8 \propto e^{-\alpha^2/2}$.

The physical interpretation is direct: the branch cut is fed by the low-frequency content of the perturbation near $\omega = 0$. A non-oscillatory Gaussian, $\nu = 0$, therefore excites the tail efficiently, whereas an oscillatory Gaussian shifts its spectrum toward $\omega \simeq \pm \nu$ and leaves only exponentially small support near zero frequency. In the far-zone regime,

$$\mathcal{A}_7 = \frac{|a_7(\nu)|}{|a_7(0)|} = e^{-\alpha^2/2} \frac{|\mathcal{B}_v^2(\alpha; r_0, \sigma)|}{|\mathcal{B}_v^2(0; r_0, \sigma)|} \simeq e^{-\alpha^2/2}, \quad (5.13)$$

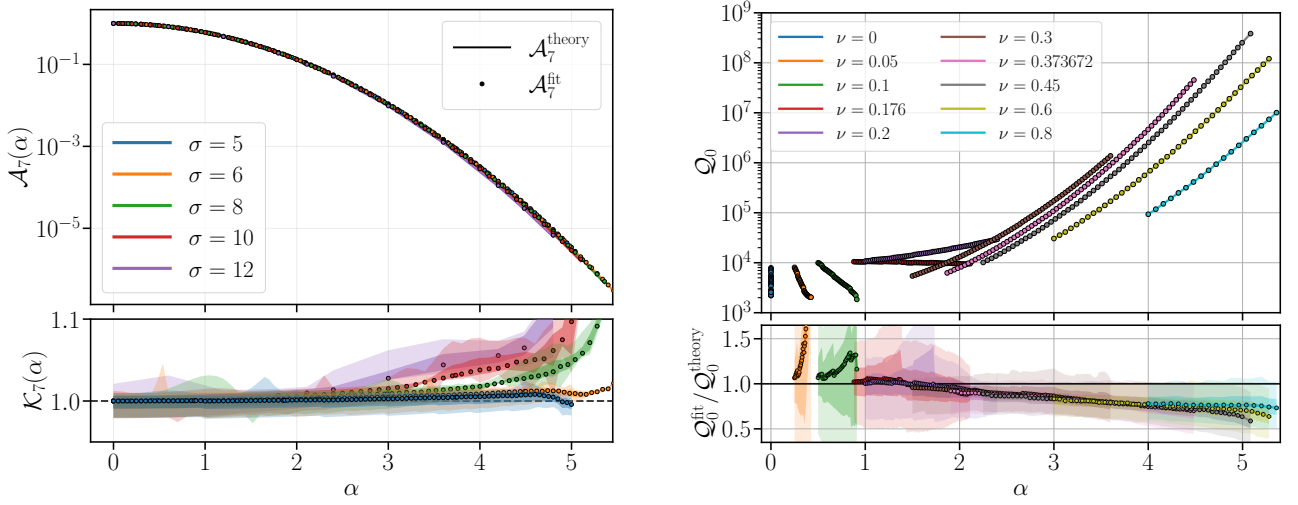


Figure 5.6: Numerical validation of the spectral suppression mechanism. Left: spectral dependence of the leading tail coefficient, with $\mathcal{K}_7 \equiv \mathcal{A}_7^{fit} / \mathcal{A}_7^{theory}$. Right: growth of the QNM-to-tail ratio as the source becomes more frequency selective. Adapted from Ref. [32].

with the same conclusion for a_8 . The factors $\mathcal{B}_v^\ell$ and $\mathcal{B}_\psi^\ell$ encode subleading radial corrections, which vary slowly compared with the dominant exponential factor.

This links the tail analysis with the excitation landscape of the previous sections and Ref. [23]: the same parameter $\alpha = \sigma\nu$ controls both the transition from broadband to frequency-selective QNM excitation and the depletion of low-frequency support feeding the branch cut. For $\alpha \lesssim 1$ the tail can remain visible, while for $\alpha \gg 1$ it is exponentially suppressed even if the QNM response stays large near ω_n^R .

The relative visibility of both sectors follows by combining the asymptotic overlap T_n^{asy} of Eq. (3.16) with an estimate of the tail amplitude around its onset time:

$$Q_n \equiv \frac{|C_n^{QNM}|}{|C^{tail}|} \propto \left| \frac{B_n \omega_n}{G_\ell^{tail}} \right| \sigma \exp \left[\frac{\alpha^2}{2} + \frac{\sigma^2}{2} \left(-(\omega_n^R - \nu)^2 + (\omega_n^I)^2 \right) \right], \quad (5.14)$$

where G_ℓ^{tail} absorbs the a_7 prefactors and the tail-onset scaling. The exponential trend changes sign when $\nu^2 - (\omega_n^R - \nu)^2 + (\omega_n^I)^2 = 0$, giving $\nu_c \simeq 0.176$ for the fundamental $\ell = 2$ mode and marking the transition from tail-favoured to QNM-favoured behaviour.

Figure 5.6 shows both checks. The left panel supports the predicted $e^{-\alpha^2/2}$ suppression of the leading tail coefficient, while the right panel shows the change of trend of the QNM-to-tail ratio around ν_c . For $\alpha \gtrsim 3$, the comparison with theory mildly deteriorates because the exponentially suppressed tail becomes harder to extract and more sensitive to fitting windows, residual contamination, and the subleading $a_8 t^{-8}$ term. This mechanism was tested in Ref. [32].

This explains perturbatively why non-oscillatory Gaussian data naturally show Price-law tails, whereas oscillatory, merger-like sources can be strongly QNM-dominated. Conversely, configurations with enhanced low-frequency content, such as eccentric or head-on mergers, are expected to display stronger tails, in agreement with recent perturbative and nonlinear studies [33–35].

Conclusions

In this work we have studied the response of a Schwarzschild black hole to localized perturbations from the point of view of the spectral content of the initial data. Using the Green-function formulation, we separated the two singular structures that control the response after the prompt transient: the quasinormal-mode poles, which govern the ringdown, and the branch cut, which generates the late-time power-law tail. This decomposition shows that the observed signal is determined not only by the spectrum of the background spacetime, but also by how the initial data project onto these different sectors of the Green function.

The first main result is that quasinormal-mode excitation can be understood as a spectral-overlap problem. For localized Gaussian perturbations, the excitation coefficient separates into a spacetime-dependent excitation factor and a source-dependent overlap. In the far-field regime this overlap reduces to a weighted Fourier transform of the initial data evaluated near the QNM frequency. The black hole therefore acts as a resonant spectral filter: frequencies contained in the perturbation efficiently excite the corresponding modes, while frequencies outside its spectral support are suppressed. The parameters σ , ν , and r_0 control, respectively, the bandwidth, carrier frequency, and spatial location of the perturbation.

The second main result is that the same spectral picture also explains the suppression of late-time tails. The branch-cut contribution is controlled by the low-frequency content of the perturbation, close to $\omega = 0$. For oscillatory Gaussian initial data this low-frequency projection is suppressed by the dimensionless parameter $\alpha = \sigma\nu$. Using Leaver's late-time expansion, the leading tail coefficients were shown to satisfy $a_7, a_8 \propto e^{-\alpha^2/2}$, up to subleading radial corrections. Thus, non-oscillatory Gaussian data naturally produce visible Price-law tails, whereas oscillatory, merger-like sources can be strongly QNM dominated. The same parameter that makes the QNM response frequency selective also depletes the low-frequency branch-cut projection.

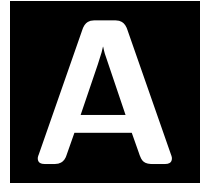
The numerical evolutions support both conclusions. The extracted QNM amplitudes reproduce the predicted excitation landscape, and the late-time fits recover the expected spectral suppression of the tail. The present work, however, remains a controlled perturbative study: it is restricted to linear Regge–Wheeler perturbations of a non-rotating Schwarzschild black hole, with prescribed initial data rather than a source generated self-consistently by a binary merger. Effects such as spin, mode mixing, nonlinearities, gauge choices and the detailed dynamics of the merger are therefore not included.

The natural next step is to extend the spectral-filtering picture to Kerr black holes, where astrophysical remnant black holes are more realistically described and the mode structure is richer. A further step would be to connect this perturbative description with numerical-relativity simulations by identifying the effective spectral content of the merger-generated perturbation. In that direction, the mechanism studied here could provide a bridge between the clean Green-function interpretation of black-hole perturbation theory and the mode amplitudes observed in fully nonlinear binary black-hole waveforms.

Bibliography

- [1] A. Einstein. Die feldgleichungen der gravitation. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 844–847, January 1915. URL: <https://ui.adsabs.harvard.edu/abs/1915SPAW.....844E>.
- [2] A. Einstein. Näherungsweise integration der feldgleichungen der gravitation. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften (Berlin)*, pages 688–696, January 1916. URL: <https://ui.adsabs.harvard.edu/abs/1916SPAW.....688E>.
- [3] C. W. Misner, K. S. Thorne, and J. A. Wheeler. *Gravitation*. W. H. Freeman, San Francisco, 1973.
- [4] R. M. Wald. *General Relativity*. The University of Chicago Press, 1984.
- [5] S. M. Carroll. *Spacetime and Geometry: An Introduction to General Relativity*. Cambridge University Press, 2019.
- [6] V. P. Frolov and I. D. Novikov. *Black hole physics : basic concepts and new developments*. 1998. URL: <https://ui.adsabs.harvard.edu/abs/1998bhp...book.....F>.
- [7] M. Maggiore. *Gravitational Waves: Volume 1, Theory and Experiments*. Oxford University Press, 2008.
- [8] D. Baumann. *General Relativity Lecture Notes*. University of Amsterdam, 2019. URL: <http://cosmology.amsterdam/wp-content/uploads/2021/10/GR-Oct19.pdf>.
- [9] Dean Rickles and Cécile M. DeWitt. An expanded version of the remarks by r. p. feynman on the reality of gravitational waves. In *The Role of Gravitation in Physics: Report from the 1957 Chapel Hill Conference*. Max-Planck-Gesellschaft zur Förderung der Wissenschaften, Berlin, 2011.
- [10] Marco Di Mauro, Salvatore Esposito, and Adele Nadeo. Towards detecting gravitational waves: a contribution by richard feynman. 2021. [arXiv:2111.00330](#).
- [11] R. A. Hulse and J. H. Taylor. Discovery of a pulsar in a binary system. *The Astrophysical Journal Letters*, 195:L51–L53, January 1975. URL: <https://ui.adsabs.harvard.edu/abs/1975ApJ...195L..51H>, doi:10.1086/181708.
- [12] B. P. Abbott et al. (LIGO Scientific Collaboration and Virgo Collaboration). Observation of gravitational waves from a binary black hole merger. *Physical Review Letters*, 116(6):061102, February 2016. [arXiv:1602.03837](#), doi:10.1103/physrevlett.116.061102.
- [13] Emanuele Berti, Vitor Cardoso, and Andrei O. Starinets. Quasinormal modes of black holes and black branes. *Classical and Quantum Gravity*, 26:163001, 2009. [arXiv:0905.2975](#), doi:10.1088/0264-9381/26/16/163001.
- [14] Luciano Rezzolla, Enrico Barausse, Ernst Nils Dornband, Denis Pollney, Christian Reisswig, Jennifer Seiler, and Sascha Husa. On the final spin from the coalescence of two black holes. *Physical Review D*, 78:044002, 2008. [arXiv:0712.3541](#), doi:10.1103/PhysRevD.78.044002.
- [15] M. Bailes et al. Gravitational-wave physics and astronomy in the 2020s and 2030s. *Nature Reviews Physics*, 3(5):344–366, January 2021. doi:10.1038/s42254-021-00303-8.
- [16] K. Schwarzschild. Über das gravitationsfeld eines massenpunktes nach der einsteinschen theorie. *Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften*, pages 189–196, January 1916. URL: <https://ui.adsabs.harvard.edu/abs/1916SPAW.....189S>.
- [17] Vitor Cardoso, Alex S. Miranda, Emanuele Berti, Helvi Witek, and Vilson T. Zanchin. Geodesic stability, lyapunov exponents, and quasinormal modes. *Physical Review D*, 79(6):064016, 2009. [arXiv:0812.1806](#), doi:10.1103/PhysRevD.79.064016.
- [18] Karl Martel and Eric Poisson. Gravitational perturbations of the schwarzschild spacetime: A practical covariant and gauge-invariant formalism. *Physical Review D*, 71:104003, 2005. [arXiv:gr-qc/0502028](#), doi:10.1103/PhysRevD.71.104003.
- [19] Alessandro Nagar and Luciano Rezzolla. Gauge-invariant non-spherical metric perturbations of schwarzschild black-hole spacetimes. *Classical and Quantum Gravity*, 22:R167, 2005. Erratum: *Classical and Quantum Gravity* 23, 4297 (2006). [arXiv:gr-qc/0502064](#), doi:10.1088/0264-9381/22/16/R01.
- [20] Edward W. Leaver. Spectral decomposition of the perturbation response of the schwarzschild geometry. *Physical Review D*, 34:384–408, 1986. doi:10.1103/PhysRevD.34.384.
- [21] Kei-ichiro Kubota and Hayato Motohashi. Resonance in black hole ringdown: Benchmarking quasinormal mode excitation and extraction. *Physical Review D*, 113:043053, 2026. [arXiv:2509.06411](#).
- [22] Nils Andersson. Evolving test fields in a black-hole geometry. *Physical Review D*, 55:468–479, 1997. [arXiv:gr-qc/9607064](#), doi:10.1103/PhysRevD.55.468.
- [23] Alejandro Svyatkovskyy Kholyavka, Jose Antonio León Vega, Samuel Gómez Gómez, Xisco Jiménez Forteza, and Sayak Datta. Shaping black hole resonances i. black hole ringdown as a spectral filtering process, 2026. URL: <https://arxiv.org/abs/2605.24704>, [arXiv:2605.24704](#).
- [24] E. W. Leaver. An analytic representation for the quasi-normal modes of kerr black holes. *Proceedings of the Royal Society of London. A*, 402:285–298, 1985. doi:10.1098/rspa.1985.0119.
- [25] Leo C. Stein. qnm: A python package for calculating kerr quasinormal modes, separation constants, and spherical-spheroidal mixing coefficients. *Journal of Open Source Software*, 4(42):1683, 2019. doi:10.21105/joss.01683.

- [26] Emanuele Berti and Vitor Cardoso. Quasinormal ringing of Kerr black holes: The excitation factors. *Physical Review D*, 74:104020, 2006. [arXiv:gr-qc/0605118](#), [doi:10.1103/PhysRevD.74.104020](#).
- [27] Emanuele Berti. Ringdown. <https://pages.jh.edu/eberti2/ringdown/>. Online data for black-hole quasinormal-mode frequencies, excitation factors, and mixing coefficients. Accessed: 2026-05-11.
- [28] Richard H. Price. Nonspherical perturbations of relativistic gravitational collapse. i. scalar and gravitational perturbations. *Physical Review D*, 5:2419–2438, 1972. [doi:10.1103/PhysRevD.5.2419](#).
- [29] E. S. C. Ching, P. T. Leung, W. M. Suen, and K. Young. Quasinormal mode expansion for linearized waves in gravitational systems. *Physical Review Letters*, 74:4588–4591, 1995. [doi:10.1103/PhysRevLett.74.4588](#).
- [30] E. S. C. Ching, P. T. Leung, W. M. Suen, and K. Young. Wave propagation in gravitational systems: Late time behavior. *Physical Review D*, 52:2118–2132, 1995. [arXiv:gr-qc/9507035](#), [doi:10.1103/PhysRevD.52.2118](#).
- [31] Carsten Gundlach, Richard H. Price, and Jorge Pullin. Late-time behavior of stellar collapse and explosions. i. linearized perturbations. *Physical Review D*, 49:883–889, 1994. [arXiv:gr-qc/9307009](#), [doi:10.1103/PhysRevD.49.883](#).
- [32] Jose Antonio León Vega, Alejandro Svyatkovskyy Kholyavka, Sayak Datta, and Xisco Jiménez Forteza. Spectral suppression of black hole ringdown tails, 2026. URL: <https://arxiv.org/abs/2606.02146>, [arXiv:2606.02146](#).
- [33] Marina De Amicis, Simone Albanesi, and Gregorio Carullo. Inspiral-inherited ringdown tails. *Physical Review D*, 110(10):104005, 2024. [arXiv:2406.17018](#), [doi:10.1103/PhysRevD.110.104005](#).
- [34] Tousif Islam, Guglielmo Faggioli, Gaurav Khanna, Scott E. Field, Maarten van de Meent, and Alessandra Buonanno. Phenomenology and origin of late-time tails in eccentric binary black hole mergers. *Physical Review D*, 112(2):024061, 2025. [arXiv:2407.04682](#), [doi:10.1103/191t-5svc](#).
- [35] Marina De Amicis, Hannes R. Rüter, Gregorio Carullo, Simone Albanesi, C. Melize Ferrus, Keefe Mitman, Leo C. Stein, Vitor Cardoso, Sebastiano Bernuzzi, Michael Boyle, Nils Deppe, Lawrence E. Kidder, Jordan Moxon, Alessandro Nagar, Kyle C. Nelli, Harald P. Pfeiffer, Mark A. Scheel, William Throwe, Nils L. Vu, and Anil Zenginoğlu. Late-time tails in nonlinear evolutions of merging black holes. *Physical Review Letters*, 135(17):171401, 2025. [arXiv:2412.06887](#), [doi:10.1103/2brx-xnyr](#).
- [36] Emanuele Berti, Vitor Cardoso, and Clifford M. Will. On gravitational-wave spectroscopy of massive black holes with the space interferometer LISA. *Phys. Rev. D*, 73:064030, 2006. [arXiv:gr-qc/0512160](#), [doi:10.1103/PhysRevD.73.064030](#).
- [37] Lionel London, Deirdre Shoemaker, and James Healy. Modeling ringdown: Beyond the fundamental quasinormal modes. *Phys. Rev. D*, 90:124032, 2014. [arXiv:1404.3197](#), [doi:10.1103/PhysRevD.90.124032](#).
- [38] Matthew Giesler, Maximiliano Isi, Mark A. Scheel, and Saul A. Teukolsky. Black Hole Ringdown: The Importance of Overtones. *Phys. Rev. X*, 9:041060, 2019. [arXiv:1903.08284](#), [doi:10.1103/PhysRevX.9.041060](#).
- [39] Swetha Bhagwat, Maria Okounkova, Stefan W. Ballmer, Duncan A. Brown, Matthew Giesler, Mark A. Scheel, and Saul A. Teukolsky. On choosing the start time of binary black hole ringdowns. *Phys. Rev. D*, 97:104065, 2018. [arXiv:1711.00926](#), [doi:10.1103/PhysRevD.97.104065](#).
- [40] Xisco Jimenez Forteza and Pierre Mourier. High-overtone fits to numerical relativity ringdowns: beyond the dismissed $n=8$ special tone. *Phys. Rev. D*, 104:124072, 2021. [arXiv:2107.11829](#), [doi:10.1103/PhysRevD.104.124072](#).
- [41] Vishal Baibhav, Mark Ho-Yeuk Cheung, Emanuele Berti, Vitor Cardoso, Gregorio Carullo, Roberto Cotesta, Walter Del Pozzo, and Francisco Duque. Agnostic black hole spectroscopy: Quasinormal mode content of numerical relativity waveforms and limits of validity of linear perturbation theory. *Phys. Rev. D*, 108:104020, 2023. [arXiv:2302.03050](#), [doi:10.1103/PhysRevD.108.104020](#).
- [42] Charles L. Lawson and Richard J. Hanson. *Solving Least Squares Problems*, volume 15 of *Classics in Applied Mathematics*. Society for Industrial and Applied Mathematics, Philadelphia, PA, 1995. [doi:10.1137/1.9781611971217](#).
- [43] Gene H. Golub and Charles F. Van Loan. *Matrix Computations*. Johns Hopkins University Press, Baltimore, MD, 4 edition, 2013.



Complementary information

A.1 List of publications derived from this work

1. Jose Antonio León Vega, Alejandro Svyatkovskyy Kholyavka, Sayak Datta, and Xisco Jiménez Forteza. *Spectral suppression of black hole ringdown tails*. arXiv:2606.02146 [gr-qc], 2026. URL: <https://arxiv.org/abs/2606.02146>.
2. Alejandro Svyatkovskyy Kholyavka, Jose Antonio León Vega, Samuel Gómez Gómez, Xisco Jiménez Forteza, and Sayak Datta. *Shaping black hole resonances I. Black hole ringdown as a spectral filtering process*. arXiv:2605.24704 [gr-qc], 2026. URL: <https://arxiv.org/abs/2605.24704>.

A.2 List of acronyms

BH	Black Hole
CFL	Courant–Friedrichs–Lewy
GR	General Relativity
GW	Gravitational Wave
NR	Numerical Relativity
QNM	Quasi-normal Mode
RW	Regge–Wheeler
TT	Transverse–Traceless
LIGO	Laser Interferometer Gravitational-Wave Observatory

Details of the Green-function construction

This appendix provides the technical steps behind the Green-function construction used in Chapter 2. We derive the frequency-domain source term, the matching formula for the Green function, the residue of a simple quasinormal pole, and the reality symmetry that allows the final waveform to be written as a real part.

B.1 Laplace transform and source term

Starting from

$$\left[\frac{\partial^2}{\partial t^2} - \frac{\partial^2}{\partial r_*^2} + V(r) \right] \Psi(t, r_*) = \mathcal{S}(t, r_*), \quad (\text{B.1})$$

we introduce the unilateral Laplace transform

$$\tilde{\Psi}(s, r_*) = \int_0^\infty e^{-st} \Psi(t, r_*) dt, \quad \text{Re}(s) > 0, \quad (\text{B.2})$$

and define

$$f(r_*) = \Psi(0, r_*), \quad g(r_*) = \partial_t \Psi(0, r_*). \quad (\text{B.3})$$

The Laplace transform of the second time derivative is

$$\mathcal{L}[\partial_t^2 \Psi] = s^2 \tilde{\Psi}(s, r_*) - s f(r_*) - g(r_*). \quad (\text{B.4})$$

Therefore Eq. (B.1) becomes

$$-\frac{d^2 \tilde{\Psi}}{dr_*^2} + [s^2 + V(r)] \tilde{\Psi} = \tilde{\mathcal{S}}(s, r_*) + s f(r_*) + g(r_*). \quad (\text{B.5})$$

Using

$$s = -i\omega, \quad \hat{\Psi}(\omega, r_*) = \tilde{\Psi}(-i\omega, r_*), \quad (\text{B.6})$$

and multiplying by -1 , one obtains

$$\frac{d^2 \hat{\Psi}}{dr_*^2} + [\omega^2 - V(r)] \hat{\Psi} = I(\omega, r_*), \quad (\text{B.7})$$

where

$$I(\omega, r_*) = i\omega f(r_*) - g(r_*) - \tilde{\mathcal{S}}(-i\omega, r_*). \quad (\text{B.8})$$

In the source-free case this reduces to

$$I(\omega, r_*) = i\omega f(r_*) - g(r_*). \quad (\text{B.9})$$

B.2 Matching construction of the Green function

The Green function is defined by

$$\left[\frac{d^2}{dr_*^2} + \omega^2 - V(r) \right] \hat{G}(r_*, r'_*; \omega) = \delta(r_* - r'_*). \quad (\text{B.10})$$

For $r_* \neq r'_*$, the delta distribution vanishes and $\hat{G}$ solves the homogeneous equation. The radiative boundary conditions then force the solution to be proportional to ψ_{in} on the left of the source point and to ψ_{up} on the right:

$$\hat{G}(r_*, r'_*; \omega) = \begin{cases} A(r'_*, \omega) \psi_{\text{in}}(r_*, \omega), & r_* < r'_*, \\ B(r'_*, \omega) \psi_{\text{up}}(r_*, \omega), & r_* > r'_*. \end{cases} \quad (\text{B.11})$$

Continuity at $r_* = r'_*$ gives

$$A(r'_*, \omega) \psi_{\text{in}}(r'_*, \omega) = B(r'_*, \omega) \psi_{\text{up}}(r'_*, \omega). \quad (\text{B.12})$$

Integrating Eq. (B.10) across an infinitesimal interval around r'_* gives the jump condition

$$\left. \frac{\partial \hat{G}}{\partial r_*} \right|_{r'_*+} - \left. \frac{\partial \hat{G}}{\partial r_*} \right|_{r'_*-} = 1. \quad (\text{B.13})$$

Substituting Eq. (B.11) into the previous two equations and solving for A and B , one finds

$$A(r'_*, \omega) = \frac{\psi_{\text{up}}(r'_*, \omega)}{W(\omega)}, \quad B(r'_*, \omega) = \frac{\psi_{\text{in}}(r'_*, \omega)}{W(\omega)}, \quad (\text{B.14})$$

where

$$W(\omega) = \psi_{\text{in}} \psi'_{\text{up}} - \psi_{\text{up}} \psi'_{\text{in}} \quad (\text{B.15})$$

is the Wronskian. Since both functions solve

$$\psi'' + [\omega^2 - V(r)] \psi = 0, \quad (\text{B.16})$$

one has

$$\frac{dW}{dr_*} = \psi_{\text{in}} \psi''_{\text{up}} - \psi_{\text{up}} \psi''_{\text{in}} = 0. \quad (\text{B.17})$$

Thus W is independent of r_* , and the Green function becomes

$$\hat{G}(r_*, r'_*; \omega) = \frac{\psi_{\text{in}}(r_*, \omega) \psi_{\text{up}}(r'_*, \omega)}{W(\omega)}, \quad r_* < r'_*, \quad r_* > r'_*. \quad (\text{B.18})$$

The Wronskian can be evaluated in either asymptotic region. At spatial infinity,

$$\psi_{\text{in}} \sim A_{\text{in}} e^{-i\omega r_*} + A_{\text{out}} e^{+i\omega r_*}, \quad \psi_{\text{up}} \sim e^{+i\omega r_*}. \quad (\text{B.19})$$

Substitution into Eq. (B.15) gives

$$W(\omega) = 2i\omega A_{\text{in}}(\omega). \quad (\text{B.20})$$

Near the horizon,

$$\psi_{\text{in}} \sim e^{-i\omega r_*}, \quad \psi_{\text{up}} \sim B_{\text{in}} e^{-i\omega r_*} + B_{\text{out}} e^{+i\omega r_*}, \quad (\text{B.21})$$

which gives

$$W(\omega) = 2i\omega B_{\text{out}}(\omega). \quad (\text{B.22})$$

Therefore

$$A_{\text{in}}(\omega) = B_{\text{out}}(\omega). \quad (\text{B.23})$$

B.3 Residue calculation and pole contribution

The solution of the inhomogeneous frequency-domain problem is

$$\hat{\Psi}(r_*, \omega) = \int_{-\infty}^{+\infty} \hat{G}(r_*, r'_*; \omega) I(\omega, r'_*) dr'_*. \quad (\text{B.24})$$

Under the change $s = -i\omega$, the inverse Laplace transform can be written as an inverse transform in the complex ω -plane. For $t > 0$, the contour is closed clockwise in the lower half-plane, giving the pole contribution

$$\Psi_{\text{QNM}}(t, r_*) = -i \sum_n e^{-i\omega_n t} \text{Res}_{\omega=\omega_n} \hat{\Psi}(r_*, \omega). \quad (\text{B.25})$$

The minus sign follows from the clockwise orientation of the closed contour.

If the source term $I(\omega, r'_*)$ is analytic near the quasinormal frequencies, the residue can be taken inside the integral:

$$\text{Res}_{\omega=\omega_n} \hat{\Psi}(r_*, \omega) = \int_{-\infty}^{+\infty} I(\omega_n, r'_*) \text{Res}_{\omega=\omega_n} \hat{G}(r_*, r'_*; \omega) dr'_*. \quad (\text{B.26})$$

For a simple pole,

$$W(\omega) = (\omega - \omega_n)W'(\omega_n) + \mathcal{O}((\omega - \omega_n)^2). \quad (\text{B.27})$$

Therefore

$$\text{Res}_{\omega=\omega_n} \hat{G}(r_*, r'_*; \omega) = \frac{\psi_{\text{in}}(r_<, \omega_n) \psi_{\text{up}}(r_>, \omega_n)}{W'(\omega_n)}. \quad (\text{B.28})$$

Combining the previous equations yields

$$\Psi_{\text{QNM}}(t, r_*) = -i \sum_n e^{-i\omega_n t} \int_{-\infty}^{+\infty} I(\omega_n, r'_*) \frac{\psi_{\text{in}}(r_<, \omega_n) \psi_{\text{up}}(r_>, \omega_n)}{W'(\omega_n)} dr'_*. \quad (\text{B.29})$$

A schematic contour deformation is shown in Fig. 2.1. The pole sum gives the quasinormal contribution, while the branch cut, omitted in the figure, gives the late-time tail.

B.4 Wave-zone limit and excitation factors

For an observer in the asymptotic wave zone, $r_* \rightarrow +\infty$, and localized source or initial-data support, one has $r_> = r_*$ and $r_< = r'_*$ inside the source integral. Then Eq. (B.29) becomes

$$\Psi_{\text{QNM}}(t, r_*) = -i \sum_{\omega_n \in \mathcal{P}_-} e^{-i\omega_n t} \frac{\psi_{\text{up}}(r_*, \omega_n)}{W'(\omega_n)} \int_{-\infty}^{+\infty} \psi_{\text{in}}(r'_*, \omega_n) I(\omega_n, r'_*) dr'_*. \quad (\text{B.30})$$

Using

$$\psi_{\text{up}}(r_*, \omega_n) \sim e^{+i\omega_n r_*}, \quad u = t - r_*, \quad (\text{B.31})$$

we obtain

$$\Psi_{\text{QNM}}(u) = -i \sum_{\omega_n \in \mathcal{P}_-} e^{-i\omega_n u} \frac{1}{W'(\omega_n)} \int_{-\infty}^{+\infty} \psi_{\text{in}}(r'_*, \omega_n) I(\omega_n, r'_*) dr'_*. \quad (\text{B.32})$$

The excitation factor is defined as

$$B_n = i \frac{A_{\text{out}}(\omega_n)}{W'(\omega_n)}. \quad (\text{B.33})$$

Since

$$W(\omega) = 2i\omega A_{\text{in}}(\omega), \quad (\text{B.34})$$

differentiation gives

$$W'(\omega) = 2iA_{\text{in}}(\omega) + 2i\omega A'_{\text{in}}(\omega). \quad (\text{B.35})$$

At a quasinormal frequency $A_{\text{in}}(\omega_n) = 0$, and therefore

$$W'(\omega_n) = 2i\omega_n A'_{\text{in}}(\omega_n). \quad (\text{B.36})$$

Thus

$$B_n = \frac{A_{\text{out}}(\omega_n)}{2\omega_n A'_{\text{in}}(\omega_n)}. \quad (\text{B.37})$$

The source-dependent overlap is defined by

$$\mathcal{T}_n = \int_{-\infty}^{+\infty} \frac{\psi_{\text{in}}(r'_*, \omega_n) I(\omega_n, r'_*)}{A_{\text{out}}(\omega_n)} dr'_*, \quad (\text{B.38})$$

so that

$$\mathcal{C}_n = B_n \mathcal{T}_n. \quad (\text{B.39})$$

Substitution into Eq. (B.32) gives

$$\Psi_{\text{QNM}}(u) = - \sum_{\omega_n \in \mathcal{P}_-} \mathcal{C}_n e^{-i\omega_n u}. \quad (\text{B.40})$$

B.5 Reality symmetry of the pole pairs

For a real potential $V(r)$ and real initial data or source, the frequency-domain solution satisfies

$$\hat{\Psi}(-\omega^*, r_*) = \hat{\Psi}(\omega, r_*)^*. \quad (\text{B.41})$$

Suppose that $\hat{\Psi}$ has a simple pole at ω_n :

$$\hat{\Psi}(\omega, r_*) = \frac{R_n(r_*)}{\omega - \omega_n} + \hat{\Psi}_{\text{reg}}(\omega, r_*), \quad (\text{B.42})$$

where $\hat{\Psi}_{\text{reg}}$ is regular at ω_n . Then Eq. (B.41) implies the existence of a reflected pole at

$$\bar{\omega}_n = -\omega_n^*. \quad (\text{B.43})$$

The lower-half-plane poles therefore appear in pairs

$$\omega_n = \omega_n^{\text{R}} - i\omega_n^{\text{I}}, \quad \bar{\omega}_n = -\omega_n^{\text{R}} - i\omega_n^{\text{I}}, \quad \omega_n^{\text{I}} > 0. \quad (\text{B.44})$$

The corresponding excitation coefficients satisfy

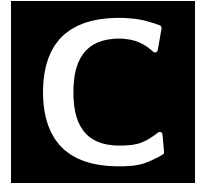
$$\mathcal{C}_{\bar{n}} = \mathcal{C}_n^*. \quad (\text{B.45})$$

Hence the contribution from one reflected pair is

$$- [\mathcal{C}_n e^{-i\omega_n u} + \mathcal{C}_n^* e^{+i\omega_n^* u}] = -2 \operatorname{Re} [\mathcal{C}_n e^{-i\omega_n u}]. \quad (\text{B.46})$$

Summing only over the representatives with positive real part gives

$$\Psi_{\text{QNM}}(u) = -2 \operatorname{Re} \sum_{\operatorname{Re}(\omega_n) > 0} \mathcal{C}_n e^{-i\omega_n u}. \quad (\text{B.47})$$



QNM-amplitude extraction from time-domain waveforms

This appendix gives the operational details of the time-domain extraction used in Section 5.3. The purpose is to define a numerical estimator for the fundamental QNM coefficient which can be compared consistently with the asymptotic and Leaver frequency-domain predictions.

The extraction strategy follows the standard ringdown practice of modelling the signal as a finite sum of damped sinusoids with QNM frequencies fixed by the background spacetime. Similar time-domain fitting strategies are widely used in black-hole spectroscopy and in numerical-relativity ringdown analyses [36–38]. In particular, the choice of fitting window, the inclusion of overtones, and the stability of the extracted amplitudes are known sources of systematic uncertainty [39–41]. Here we adapt this idea to a controlled Regge–Wheeler evolution: the frequencies are kept fixed, the local complex amplitudes are fitted in sliding windows, and the reported fundamental coefficient is obtained by averaging the complex local estimates over a selected ringdown interval [23].

C.1 Reference-time convention

The waveform is extracted at a fixed observer radius r_*^{obs} . We use the retarded time

$$u = t - r_*^{obs}, \quad (\text{C.1})$$

and introduce the shifted time coordinate

$$\tau = u - u_{ref}. \quad (\text{C.2})$$

The value of u_{ref} is chosen according to the scan. For scans at fixed initial position, $u_{ref} = r_0$. When the initial position itself is varied, the same prescription is applied to each waveform separately, so that the fitted coefficient is always referred to the time origin associated with the corresponding initial pulse.

With this convention, the numerical fitting model for the fundamental mode is

$$\Psi(u) \simeq \text{Re} \left[C_0^{num}(\tau_0) e^{-i\omega_0 \tau} \right], \quad \tau = u - u_{ref}. \quad (\text{C.3})$$

The quantity τ_0 labels the starting point of the fitting window in the shifted time coordinate. It is not a new physical time, but only a parameter controlling where the local fit is performed.

C.2 Locally rephased least-squares fit

Directly fitting Eq. (C.3) over long time intervals can be numerically ill-conditioned, because the exponentially damped QNM basis changes strongly across the full waveform. For this reason, each window is fitted in a locally rephased basis.

Let a fitting window be

$$W(\tau_0) = [\tau_0, \tau_0 + T_{win}], \quad (\text{C.4})$$

and denote its centre by

$$\tau_w = \tau_0 + \frac{T_{win}}{2}. \quad (\text{C.5})$$

Inside this window we fit

$$\Psi(u) \simeq \text{Re} \left[D_0(\tau_0) e^{-i\omega_0(\tau - \tau_w)} \right]. \quad (\text{C.6})$$

The fitted coefficient $D_0(\tau_0)$ is then transformed back to the global reference convention through

$$C_0^{num}(\tau_0) = D_0(\tau_0) e^{i\omega_0 \tau_w}. \quad (\text{C.7})$$

This is only a change of numerical basis: the final coefficient is the one appearing in Eq. (C.3).

More generally, for a set of fitted modes $\mathcal{N}$, the local model can be written as

$$\Psi(u) \simeq \sum_{n \in \mathcal{N}} \text{Re} \left[D_n(\tau_0) e^{-i\omega_n(\tau - \tau_w)} \right] + P_q(\tau - \tau_w), \quad (\text{C.8})$$

where P_q is an optional low-order polynomial trend. In the scans shown in Section 5.3, the reported quantity is the fundamental coefficient C_0^{num} . Higher modes are mainly used as a diagnostic of the stability of the extraction.

The fit is linear in the real and imaginary parts of the amplitudes. We write

$$D_n = a_n + ib_n, \quad X_n(\tau) = e^{-i\omega_n(\tau - \tau_w)}. \quad (\text{C.9})$$

Then

$$\text{Re} [D_n X_n] = a_n \text{Re}(X_n) - b_n \text{Im}(X_n). \quad (\text{C.10})$$

Thus each complex amplitude contributes two real columns to the design matrix. The fitted parameters are obtained by a standard real linear least-squares problem [42, 43],

$$\chi^2(\tau_0) = \sum_{\tau_j \in W(\tau_0)} \left[\Psi_j - \sum_{n \in \mathcal{N}} (a_n \text{Re}(X_{n,j}) - b_n \text{Im}(X_{n,j})) - P_q(\tau_j - \tau_w) \right]^2. \quad (\text{C.11})$$

This real formulation is equivalent to fitting the complex amplitudes directly, but it avoids introducing a complex-valued regression problem for a real waveform. The QNM frequencies are not fitted: they are fixed to their Schwarzschild values. Therefore the only fitted quantities in each local window are the amplitudes, and possibly the nuisance coefficients of the low-order trend.

C.3 Ringdown-window amplitude and sensitivity bands

The sliding-window fit produces, for each waveform, a sequence of complex local estimates

$$C_0^{num}(\tau_0).$$

For a fixed reference ringdown interval I_{RD} , the numerical amplitude shown in the parameter scans is defined as

$$|C_0^{num}| = \left| \langle C_0^{num}(\tau_0) \rangle_{\tau_0 \in I_{RD}} \right|. \quad (\text{C.12})$$

Thus the complex amplitudes are averaged before taking the modulus. This keeps the phase convention of the fitted QNM coefficient and avoids using $\langle |C_0^{num}(\tau_0)| \rangle$ as the plotted estimator.

The shaded bands in the parameter scans quantify the sensitivity of the extraction to the choice of ringdown interval. The choice of the start time or fitting interval is a known issue in ringdown analyses, especially when overtones or non-ringdown components are present [38–41]. In the present

work, the bands are not statistical confidence intervals for the theoretical curves. They are deterministic robustness diagnostics associated with the extraction procedure.

For each admissible interval W in the shifted time coordinate, we compute

$$A_W = \left| \langle C_0^{num}(\tau_0) \rangle_{\tau_0 \in W} \right|. \quad (\text{C.13})$$

The set of values $\{A_W\}$ defines the distribution induced by changing the extraction window. The central curve is obtained from the fixed reference interval I_{RD} , while the bands are obtained from percentiles of this admissible-window distribution.

The reference interval I_{RD} is fixed for each scan and is always defined in the shifted coordinate $\tau = u - u_{ref}$. In the numerical implementation, the endpoints of the admissible intervals W are varied over a grid in τ . Intervals shorter than the chosen minimum duration, or containing too few local fits, are discarded. The inner band corresponds to the central 16–84 percentile range of $\{A_W\}$, while the outer band corresponds to the central 0.135–99.865 percentile range:

$$A_{lo}^{(68)}, A_{hi}^{(68)} = \text{percentile}_{16, 84}(\{A_W\}), \quad (\text{C.14})$$

$$A_{lo}^{(99.7)}, A_{hi}^{(99.7)} = \text{percentile}_{0.135, 99.865}(\{A_W\}). \quad (\text{C.15})$$

A narrow band indicates that the inferred amplitude is insensitive to moderate changes of the ringdown window. A broad band indicates stronger contamination from prompt response, overtones, late-time tails, or window-selection effects.

This procedure is applied independently to each waveform in the parameter scan. Therefore the bands in the ν - and σ -dependent plots do not come from a single representative plateau, but from repeating the same extraction and window-variation procedure for every value of the scanned parameter.

C.4 Conversion of theoretical amplitudes

The theoretical amplitudes must be expressed in the same reference-time convention as the numerical fit. We use here the reduced coefficient C_0 defined in Chapter 3, where the mirror-pole factor and the translation factor associated with r_0 have already been absorbed. With this convention, the fundamental QNM contribution is written as

$$\Psi_{QNM}(u) = -\text{Re} \left[C_0 e^{-i\omega_0(u-r_0)} \right]. \quad (\text{C.16})$$

The numerical fit is performed with the common reference time u_{ref} ,

$$\Psi(u) \simeq \text{Re} \left[C_0^{common} e^{-i\omega_0(u-u_{ref})} \right]. \quad (\text{C.17})$$

Equating Eqs. (C.16) and (C.17) gives

$$C_0^{th, common} = -C_0 e^{-i\omega_0(u_{ref}-r_0)}. \quad (\text{C.18})$$

For scans at fixed initial position, $u_{ref} = r_0$, and therefore

$$C_0^{th, common} = -C_0. \quad (\text{C.19})$$

The overall minus sign is irrelevant for amplitude-only comparisons, but it is kept here to make the phase convention explicit.



Fourth-order finite-difference stencils

This appendix collects the finite-difference formulas used in the time-domain code. The grid is uniform,

$$r_{*,j} = r_{*,\min} + jh, \quad u_j(t) = u(t, r_{*,j}),$$

and all expressions below are fourth-order accurate.

D.1 First derivative

In the bulk, the first derivative is approximated by the centered five-point stencil

$$(\partial_{r_*} u)_j = \frac{-u_{j+2} + 8u_{j+1} - 8u_{j-1} + u_{j-2}}{12h} + \mathcal{O}(h^4), \quad 2 \leq j \leq N-3. \quad (\text{D.1})$$

Near the left boundary, the one-sided closures are

$$(\partial_{r_*} u)_0 = \frac{-25u_0 + 48u_1 - 36u_2 + 16u_3 - 3u_4}{12h} + \mathcal{O}(h^4), \quad (\text{D.2})$$

$$(\partial_{r_*} u)_1 = \frac{-3u_0 - 10u_1 + 18u_2 - 6u_3 + u_4}{12h} + \mathcal{O}(h^4). \quad (\text{D.3})$$

Near the right boundary,

$$(\partial_{r_*} u)_{N-2} = \frac{-u_{N-5} + 6u_{N-4} - 18u_{N-3} + 10u_{N-2} + 3u_{N-1}}{12h} + \mathcal{O}(h^4), \quad (\text{D.4})$$

$$(\partial_{r_*} u)_{N-1} = \frac{25u_{N-1} - 48u_{N-2} + 36u_{N-3} - 16u_{N-4} + 3u_{N-5}}{12h} + \mathcal{O}(h^4). \quad (\text{D.5})$$

These first-derivative formulas are used to impose the Sommerfeld boundary conditions and to construct the initial velocity $v(0, r_*) = \partial_{r_*} u(0, r_*)$.

D.2 Second derivative

In the bulk, the second derivative is computed with

$$(\partial_{r_*}^2 u)_j = \frac{-u_{j+2} + 16u_{j+1} - 30u_j + 16u_{j-1} - u_{j-2}}{12h^2} + \mathcal{O}(h^4), \quad 2 \leq j \leq N-3. \quad (\text{D.6})$$

At the first two grid points,

$$(\partial_{r_*}^2 u)_0 = \frac{35u_0 - 104u_1 + 114u_2 - 56u_3 + 11u_4}{12h^2} + \mathcal{O}(h^4), \quad (\text{D.7})$$

$$(\partial_{r_*}^2 u)_1 = \frac{10u_0 - 15u_1 - 4u_2 + 14u_3 - 6u_4 + u_5}{12h^2} + \mathcal{O}(h^4). \quad (\text{D.8})$$

At the last two grid points,

$$(\partial_{r_*}^2 u)_{N-2} = \frac{10u_{N-1} - 15u_{N-2} - 4u_{N-3} + 14u_{N-4} - 6u_{N-5} + u_{N-6}}{12h^2} + \mathcal{O}(h^4), \quad (\text{D.9})$$

$$(\partial_{r_*}^2 u)_{N-1} = \frac{35u_{N-1} - 104u_{N-2} + 114u_{N-3} - 56u_{N-4} + 11u_{N-5}}{12h^2} + \mathcal{O}(h^4). \quad (\text{D.10})$$

The discrete spatial operator used in the evolution is therefore

$$L_h[u]_j = (\partial_{r_*}^2 u)_j - V_j u_j, \quad V_j \equiv V(r_{*,j}). \quad (\text{D.11})$$